



The aircraft recovery problem: A systematic literature review

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ABSTRACT

Events that cause disruptions to air company flight schedules transporting passengers and cargo are commonplace in practice, and their operational and financial consequences are often relevant. The aircraft recovery problem (ARP) consists of recovering the flight schedules lost due to such events, determining new flight departure times and possible flight cancellations, as well as revising routes for different aircraft. In this study, the main characteristics considered in ARP papers to make the problem as realistic as possible are identified and a systematic literature review is carried out based on seminal studies of the problem in the 1980s. The literature papers are reviewed in terms of the ARP variants studied, the objectives chosen to be optimized and the practical constraints most often considered in real applications, as well as the networks and mathematical formulations to represent the problem and the solution approaches used to deal with it. The aim is to present an up-to-date review of the state-of-the-art of the ARP and to identify possible literature gaps and interesting opportunities for future research on this problem. It is noteworthy that, compared to the simplicity of the first ARP studies, the studies of the following decades tended to involve more complex variants of the problem as a way to become more adherent to current practical environments, requiring more elaborate formulations and solution methods to be properly addressed.

1. Introduction

An air operator is a legal entity that transports people or cargo between two different airports, i.e., areas intended for landing, take-off and aircraft movement (IAC 1223, 2000). According to Clausen et al. (2010), when an interruption in the operations of an air operator happens, so that it fails to perform one or more scheduled flights, the execution of the next scheduled flights may be compromised and, thus, adversely affect the passengers' itineraries. Regarding commercial aviation, disruptions in flight plans can cause considerable impacts on airlines. For example, prior to the recent Covid-19 pandemic, in the United States in 2018, costs resulting from such delays were estimated to have reached US\$28 billion, according to the Federal Aviation Administration (FAA) (Airlines For America, 2018). In addition to the undesirable deterioration in the level of service to users and the high financial costs, such disturbances are also a matter of concern due to their high frequency of occurrence. In Europe, according to a report by the European Organization for the Safety of Air Navigation (EUROCONTROL), also prior to the pandemic, the number of flights per day grew by 3.4% between February 2017 and February 2018 and the percentage of flights delayed at departure airports has also increased. Ground delays greater than 60 min increased from 2.9%

to 4.2% compared to February 2017, and the average delay per flight increased by 3.1 min to reach 12.6 min. The main cause of such delays was bad weather conditions (Kammoun and Rezg, 2018). Therefore, as a way of minimizing the negative impacts of such events, it is important that operators are prepared for these unforeseen events and have effective planning tools for managing interruptions.

According to Belobaba et al. (2015), when this untimely scenario occurs and most or all of the flight schedules for a certain period are lost, several decisions must be taken so that such flights are rescheduled with minimal loss to airlines and discomfort for their users. In this context, optimization approaches and other operations research techniques can be useful to assist in the decision-making process, considering the complex characteristics involved in a (generally) short period of time, characteristics and decisions that comprise the so-called *airline recovery problem*. This problem consists of three other sub-problems: the *crew recovery problem* (CRP), which essentially deals with the recovery of the crew's scale; the *passenger recovery problem* (PRP), which addresses the recovery of passenger itineraries; and the *aircraft recovery problem* (ARP), which aims to determine new departure times for flights, as well as possible flight cancellations, and review routes for different aircraft.

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The ARP, in particular, has a specific characteristic: it is usually the first sub-problem to be solved when trying to address the airline recovery problem sequentially (Belobaba et al., 2015). Given a flight schedule and a set of external and internal disruptions, the ARP consists of determining which flights to delay or cancel, and reassigning available aircraft to the flights in order to minimize the undesirable consequences of disruptions. Aircraft unavailability and failures and airport service disturbances are usually the main causes of flight disruptions, and the problem generally assumes that the crew is available to perform flights in the recovery schedule (Hassan et al., 2021). Teodorović and Guberinić (1984) was one of the first studies addressing the ARP in the 1980s. The problem was represented by a graph depicting all chronological series (chains) of flights the aircraft could execute during the course of the day. In view of the time losses experienced by passengers due to flight reassignment when one or more aircraft become unavailable, the authors devised a method to obtain the chain of flights that prioritizes the recovery of fuller flights over the emptiest ones by minimizing the total passenger delay. Few practical constraints were considered; each aircraft must not depart before the flight's original departure time and a minimum predefined interval (block and turnaround time) between the departure of every two of the aircraft's consecutive flights must be complied.

Since then, a number of studies has emerged on the topic, each one either including new particularities and exploring variations of the ARP, such as additional realistic constraints of practical applications, decisions and multiple optimization objectives, or presenting new network representations and mathematical formulations as well as alternative solution approaches to the problem. Typical recovery actions on these ARP studies include flight delays, flight cancellations, and aircraft swaps. Given the high impact that the ARP may cause on the operations of organizations with air transport activities, as well as its high frequency of occurrence, it is important to collect and analyze such studies in order to identify which aspects of a particular problem the literature already addresses, and which models and solution approaches can be used to solve it. To achieve this goal, the *systematic literature review* (SLR) method is particularly interesting (Sampaio and Mancini, 2007; Denyer et al., 2009; Jesson et al., 2011). In an SLR, evidence is integrated and presented related to a specific topic or set of related topics. It uses explicitly systematized methods for searching, critically evaluating and summarizing the knowledge collected, and is effective in consolidating information from a particular collection of articles.

The present review focuses on the ARP and optimization and operations research approaches proposed in the literature to address it. All studies within the criteria imposed in the review protocol that deal with the ARP were considered for systematic evaluation. In this study, an SLR is carried out on the ARP to identify which are the main characteristics considered in the reference papers of the literature, which are the main decisions involved and which models and solution methods are most used to deal with them. Several papers are reviewed and evaluated in terms of the different aspects and constraints of the problems considered, in addition to the main optimization criteria used. The review also aims to identify possible literature gaps and interesting opportunities for future research on this topic.

1.1. Previous ARP reviews in the literature

There are some previous literature review studies that compiled and analyzed the knowledge developed regarding rescheduling problems. These reviews are usually limited to a single mode of transportation, the most studied of which is on air (Filar et al., 2001; Kohl et al., 2007; Clausen et al., 2010; Hassan et al., 2021), but there are also studies on rail (Cacchiani et al., 2014), road (Eglese and Zambirinis, 2018) and maritime (Lee and Song, 2017) modes. Moreover, there are those who encompassed more than one mode of transportation.

Regarding the reviews that comprise the air mode, Filar et al. (2001) and Kohl et al. (2007) addressed the topic comprehensively. The authors reviewed the associated literature and discussed the airline disruption management, addressing the main factors that influence the instability of flight control operations. In addition, they also pointed out the main management processes that are involved in this recovery. Kohl et al. (2007) did not specifically address the ARP, nor did they discuss mathematical formulations for the problem or existing solution methods. In their reviews, Clausen et al. (2010) and Hassan et al. (2021) compared studies in the literature on airline disruption management regarding the network representations, mathematical modeling and solution methods used. Clausen et al. (2010) made such comparisons with studies focused both on ARP and CRP, while Hassan et al. (2021) directed at the works that coordinate two or more resources (aircraft, crew, passengers), thus exploring different integrations among the ARP, CRP and PRP regarding the recovery analysis.

Visentini et al. (2014) reviewed models and solution methods used in typical rescheduling problems not only of air, but also rail and road modes of transportation. Even though the present review covers only the ARP, it is worth briefly mentioning some peculiarities in other modes. For example, for road-vehicle services such as intercity bus transportation, school transportation and urban waste collection, the literature points out a tendency for tackling the associated recovery problems as a dynamic vehicle scheduling problem, either mono-depot or multi-depot. Differently from the commercial aerial mode, there are few companies or agencies that use computerized rescheduling in the occurrence of vehicle unavailability or delays. Vehicle breakdown frequently occurs on the road, which requires recovery decisions that take into account the extent of time that crew, passengers and cargo have to endure in a unattended and potentially dangerous location.

Train scheduling, on the other hand, is a difficult problem mostly because of the limited physical resources (tracks and stations) that invariably led to many operational constraints that rule train movements. Rescheduling often implies in the reallocation of trains to different track blocks when they reach a classification yard, hence altering the initial timetable, and mainly aiming the minimization of the total delay. Due to its great complexity, simulation and heuristic approaches have been proposed in order to provide good solutions in reasonable computational times. Some examples of train schedule recovery problems are found in, e.g., Luethi et al. (2009), Berger et al. (2011), Krasemann (2012) and Almodóvar and García-Ródenas (2013).

Lastly, it is noteworthy the schedule recovery in liner shipping, i.e., the mode of maritime transportation mostly used in containerized trade around the world and whose service operates on a timed schedule with a fixed port rotation (Abioye et al., 2019, 2021). In maritime transportation, the ships' original schedule is subject to uncertainties not only at sea but also at ports due to congestion in channels that give access to terminals and cargo handling at terminals. A number of mathematical models prescribes speed adjustment and skipping or swapping ports of call as possible scheduling recovery actions. Rather than addressing delays directly, the objective is often the minimization of the total cost of route service and penalties imposed to the liner shipping company as a result of delays or missed cargo connection. For more details of ship schedule recovery problems focusing on cargo transportation, see, e.g., Dulebenets et al. (2021), Elmi et al. (2022) and Li et al. (2022) and the references therein.

1.2. Contributions and organization of this study

The present study differs from these previous literature reviews on rescheduling problems mainly because it deals only with the air mode, only with the ARP and only with passenger transportation, focusing on different variants of ARP with different characteristics, formulations and solutions methods. Moreover, it uses a SLR method,

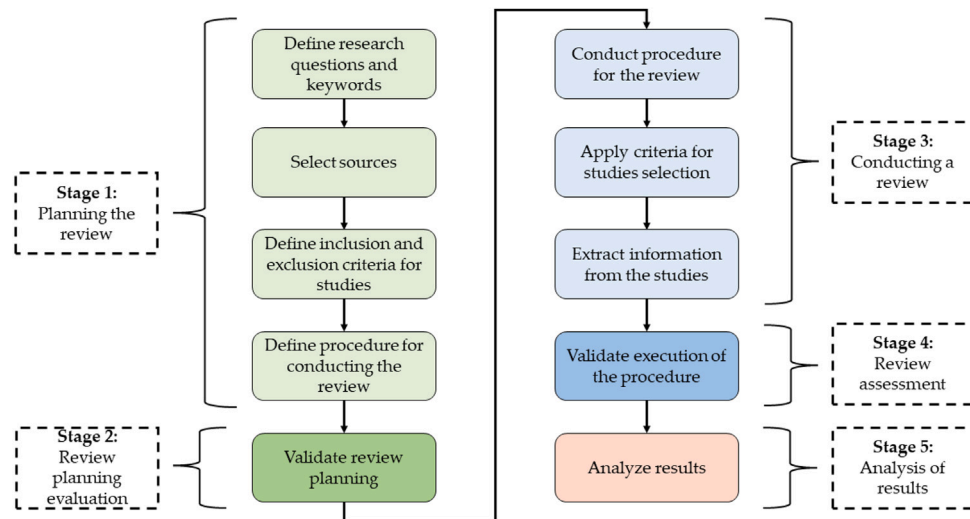


Fig. 1. Stages of the SLR.

thus providing a larger pool of evaluated studies on the ARP and whose selection and appreciation followed a method less prone to bias and subjectivity. Previous reviews of the literature on aircraft disruption management are not up to date or focus on related but more general issues than the ARP. Additionally, they do not apply a SLR method, which, in turn, offers certain advantages in terms of method rigor and results that non-systematized reviews may undergo (Biolchini et al., 2005). A SLR identifies relevant studies from the literature, evaluating their contributions and summarizing their results. It helps to answer important research questions, such as what the main gaps are between academic literature and airline practices, what trends are observed in the representation of the ARP in terms of modeling choices and analytical solution methods, how the ARP has been most often addressed in the literature, among others. Some of the contributions of this paper are: to analyze and describe some current gaps in the literature, to highlight recent trends in the literature and to point out some interesting opportunities for future research on the ARP.

This study is organized as follows: in Section 2, the steps of this SLR method are presented and discussed. In Section 3, the reviewed studies are analyzed in detail. In Section 4, the main results and insights obtained from the SLR, as well as research gaps and perspectives for future research, are highlighted. Finally, in Section 5, the conclusions of this review are presented.

2. The systematic literature review method

As previously mentioned, the SLR is an effective research method to review the literature and identify gaps and opportunities for future research on a certain topic (Sampaio and Mancini, 2007). For this review, the SLR was conducted especially using the method described in Biolchini et al. (2005) (see also Denyer et al. (2009) and Jesson et al. (2011)), as illustrated in Fig. 1. After the review planning stage (stage 1 in Fig. 1), the result is the research protocol, which describes the particularities of how the SLR was carried out to answer the proposed research questions. For its construction, insights obtained from reading other literature reviews (not necessarily systematic) were also applied, such as Uhlmann and Frazzon (2018), Clausen et al. (2010), Visentini et al. (2014) and Hassan et al. (2021).

In order to identify characteristics of the ARP already discussed in the literature, as well as the network representations, models and solution approaches, the main characteristics of the problems dealt with in the reviewed articles were listed. Since the first papers reported in the literature, some features referred in the studies include whether

the problem is single-fleet or multi-fleet, if flow control, capacity limitations and operating time windows (e.g., curfews) are considered at airports, if operating time windows are imposed to aircraft, whether real-life or fictional data are used to evaluate the proposed approaches and if there are additional features or constraints, among others. In addition to minimizing flight delays in rescheduling, other objectives were also identified in the published articles, such as minimizing exchanges of aircraft previously allocated to flights and minimizing flight cancellations, among others. The solution methods adopted by the authors are generally classified into heuristic and exact methods. Fig. 2 summarizes these characteristics of ARPs, while Fig. 3 details the search protocol of the SLR applied.

2.1. Some descriptive characteristics of the articles

After applying the protocol, several articles were eliminated from those resulting from the mere application of the research strings in the databases, including the ones that integrate the ARP with the CRP and/or the PRP, concluding the SLR with 50 articles, as shown in Fig. 4. A few other related articles not found by the SLR, which were of our prior knowledge, were also included in this review. The distribution of the publications of the articles over the years is depicted in Fig. 5. There were two moments of increase: the first between 1995 and 1997, and the second from 2012 onwards until 2022, particularly in the last years. It should be noted that articles published from 2012 onwards were not covered in almost all of the previous literature reviews on ARP referred to in Section 1.1.

Regarding the journals in which articles on the topic were published, as reported in Table 1, there is a homogeneity of titles in which few contain some publications more than the rest. However, in general terms, there is no predominance of any specific journal. Regarding the countries of the authors of the published studies, Table 2 illustrates the number of articles per country. In this table, for each study, one article was counted for each nationality of the respective authors. In the case of more than one author with the same nationality, that country was counted only once. As a result, a predominance of studies published by authors from the USA, followed by China and France, respectively, can be noted, although the seminal study appeared in the region that is now Serbia.

Group	ARP characteristic	Description
Features What aspects of the ARP are addressed in the article?	Aircraft fleet	Is the ARP single-fleet or multi-fleet?
	Airport flow control	Is there any kind of airport flow control and/or capacity limitation?
	Airport time windows	Do the airports have operating time windows?
	Aircraft time windows	Do the aircraft have operating time windows?
	Use of real data	Is the solution method applied in situations using real data?
	Other practical features	Are there additional practical features?
Objectives What are the objectives to be optimized?	Delays	Should the flight delays that must be recovered be minimized?
	Swaps	Should swapping of aircraft previously allocated to interrupted flights be minimized?
	Cancellation	Should flight cancellations be minimized?
	Other objectives	Are there other objectives in the problem?
Solution methods What are the solution methods used by the authors?	Heuristic	Are the solution methods heuristic or metaheuristic?
	Exact	Is there any optimization solver used or is an exact solution method developed?

Fig. 2. Characteristics of the ARPs investigated in the SLR articles.

Table 1

Journals in which the SLR articles were published.

Journal	Number of publications
Transportation Science	5
Transportation Planning and Technology	4
European Journal of Operational Research	3
Computers & Operations Research	3
Mathematical Problems In Engineering	3
IEEE Access	3
Transportation Research Part B-Methodological	2
Expert Systems With Applications	2
IIE Transactions	2
Asia-Pacific Journal of Operational Research	2
Journal Of Air Transport Management	2
Journal of Transportation Engineering	1
Journal Of The Operational Research Society	1
International Journal of Operational Research	1
Journal of Combinatorial Optimization	1
Transportation Research Part A: Policy and Practice	1
Journal of Heuristic	1
Journal Of Industrial And Management Optimization	1
IET Intelligent Transport Systems	1
Central European Journal of Operations Research	1
Kybernetes	1
International Journal Of Production Research	1
Studies in Informatics & Control	1
Computers & Industrial Engineering	1
International Journal Of Advanced Trends In Computer Science And Engineering	1
Sustainability	1
Chinese Journal Of Aeronautics	1
Transportation Letters	1
Discrete Dynamics In Nature And Society	1
Applied Soft Computing	1
Aerospace Science and Technology	1
Transportation Research Part E-Logistics And Transportation Review	1
International Transactions in Operational Research	1
Euro Journal on Transportation and Logistics	1
Operations Research	1

Table 2

Number of SLR articles published by country.

Country	Number of articles
USA	18
China	17
France	6
Brazil	4
Yugoslavia ^a	3
Taiwan	3
Turkey	2
Sweden	2
Australia	2
Singapore	1
Hong Kong	1
Netherlands	1
Italy	1
Republic of Korea	1
Switzerland	1
Germany	1
Denmark	1

^aThe authors at the time were affiliated to research institutions in Belgrade, the current capital of Serbia.

2.2. A first study on the ARP

As already mentioned, the seminal study on ARP is commonly attributed to [Teodorović and Guberinić \(1984\)](#). For illustration purposes and to make the problem clearer and more detailed for the reader, the pioneer ARP studied by the authors and the model proposed to represent it are briefly described below. The authors considered a situation of an airline in which one or more aircraft previously assigned to flights scheduled for a given day are out of operation at the beginning of that day. The aim is to determine a new schedule in order to minimize the total passenger delay across the entire flight network.

It is assumed that all aircraft in the fleet have the same capacity, i.e., the fleet is homogeneous (considered as a single fleet with same aircraft type and ignoring all maintenance constraints), there are no limitations regarding flow control and capacity for the airports nor time

Research question
Question: What are the characteristics and solution methods of the ARP studied in the literature?
Keywords: flight rescheduling; aircraft rescheduling; airplane rescheduling; helicopter rescheduling; airport disruption management; aircraft routing and rescheduling problem; helicopter routing problem; flight rescheduling problem; aircraft recovery problem; oil platform transport problem; rescheduling airline;
Publication years: No filters.
Source
Sources: Compendex, Scopus, Web of Science.
String:
Compendex: ((((((((((flight AND rescheduling) WN KY) OR ((aircraft AND rescheduling) WN KY)) OR ((airplane AND rescheduling) WN KY)) OR ((helicopter AND rescheduling) WN KY)) OR (("airport disruption management") WN KY)) OR (("aircraft routing and rescheduling problem") WN KY)) OR (("helicopter routing problem") WN KY)) OR (("flight rescheduling problem") WN KY)) OR (("aircraft recovery problem") WN KY)) OR (("oil platform transport problem") WN KY)) OR ((rescheduling AND airline) WN KY)) Scopus: TITLE-ABS-KEY (flight AND rescheduling) OR TITLE-ABS-KEY (aircraft AND rescheduling) OR TITLE-ABS-KEY (airplane AND rescheduling) OR TITLE-ABS-KEY (helicopter AND rescheduling) OR TITLE-ABS-KEY ("airport disruption management") OR TITLE-ABS-KEY ("aircraft routing and rescheduling problem") OR TITLE-ABS-KEY ("helicopter routing problem") OR TITLE-ABS-KEY ("flight rescheduling problem") OR TITLE-ABS-KEY ("aircraft recovery problem") OR TITLE-ABS-KEY ("oil platform transport problem") OR TITLE-ABS-KEY (rescheduling AND airline) Web of Science: TS=(flight AND rescheduling) OR TS=(aircraft AND rescheduling) OR TS=(airplane AND rescheduling) OR TS=(helicopter AND rescheduling) OR TS=("airport disruption management") OR TS=("aircraft routing and rescheduling problem") OR TS=("helicopter routing problem") OR TS=("flight rescheduling problem") OR TS=("aircraft recovery problem") OR TS=("oil platform transport problem") OR TS=(rescheduling AND airline)
Study selection
Inclusion and exclusion criteria of studies: • The text must be entirely in English, not just the title, abstract or keywords • The study must be a journal article • The article must have the ARP as its main decisions.
Information to be collected (Classification criteria): • Descriptive information of studies: year, authors and means of publication • Decisions, optimization objectives, features considered by the models and solution methods used
Procedure for study classification: 1. Apply specified string in database search engines; 2. Apply language and means of publication filters; 3. For each text, repeat the following steps, until all the articles returned by the database are evaluated with the following subjects: (a) Check if the article is published in a journal (b) Check the data provided by the database (title and abstract) if the article deals with flight recovery (c) Check through full reading if the article deals only with the ARP 4. In case of positive answers for items 3a, 3b and 3c, include the article in the SLR 5. In case there is at least one negative answer for items 3a, 3b or 3c, do not include the article and return to step 3 6. For each article, locate the information described in Figure 2 and register it.

Fig. 3. SLR protocol.

windows for the airport and aircraft operation, and the objective is to minimize flight delays in rescheduling. Let $X = \{1, 2, \dots, K\}$ be the set of flights that must be performed. For each flight $i \in X$, the following data is given: the departure time a_i , which is the previously scheduled departure time of flight i before rescheduling; and planned time b_i , when the aircraft allocated to perform flight i will be ready for some other flight. Time interval $b_i - a_i$ represents the total operating time required by flight i , that is, the instant from the moment the aircraft

engine that will perform it is started at the airport of origin until its shutdown at the airport of destination, plus the time needed to prepare that same aircraft for another flight. The number of passengers p_i for each flight i is also known.

In a new schedule, each aircraft must not depart before the original departure time of the flight. In other words, if c_i is the new scheduled departure time for flight i , then after rescheduling: $c_i \geq a_i, i \in X$. Additionally, the aircraft must be ready to perform a new flight after

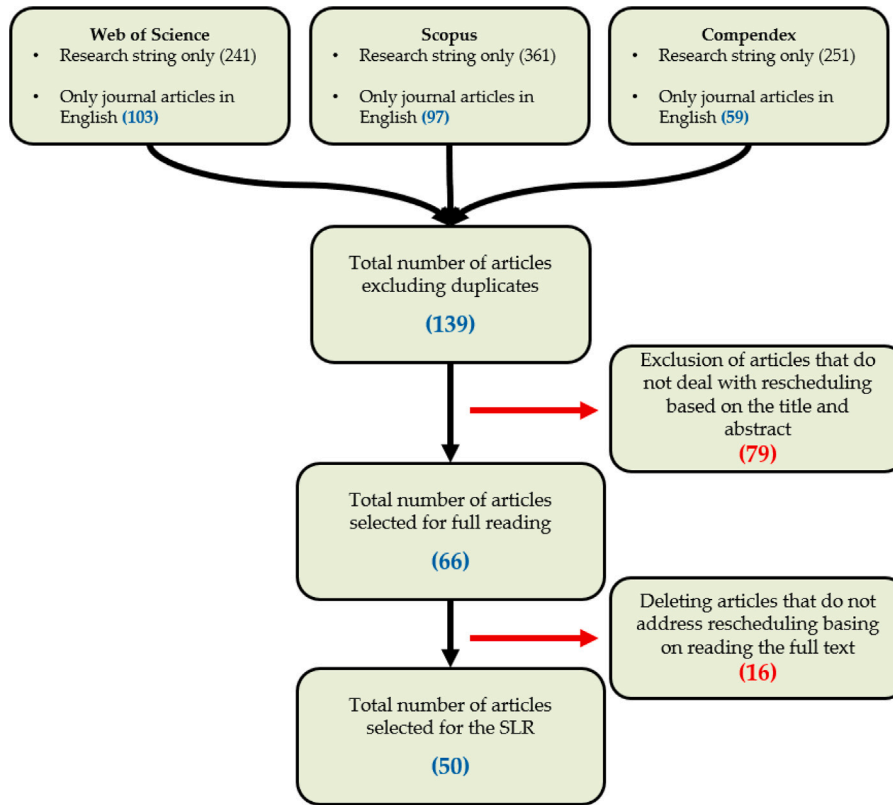


Fig. 4. Results of the SLR.

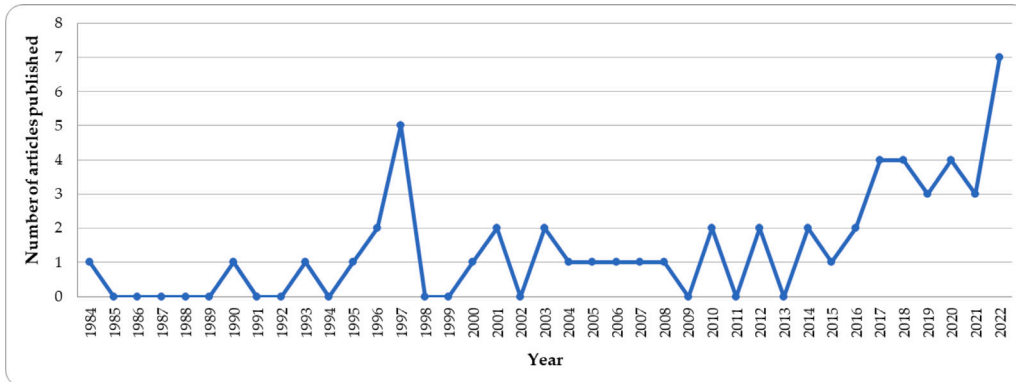


Fig. 5. Number of SLR articles published over the years.

completing the flight i , i.e., at instant d_i given by: $d_i = c_i + (b_i - a_i)$. The delay for each flight i in this new schedule is calculated as the time difference between the new scheduled departure time c_i and the original departure time a_i , i.e., $c_i - a_i$. When setting a new departure time, the number of passengers p_i on each flight must be taken into account. Therefore, the problem is to determine the new scheduled departure times c_i for the flights to minimize the total weighted delay of all flights: $W = \sum_{i \in X} p_i(c_i - a_i)$.

The following illustrative example, as shown in Teodorović and Guberinić (1984), contains 3 airports (A_1 , A_2 and A_3), $K = 8$ scheduled flights between these airports and initially 3 aircraft (a , b and c) considered identical to perform all these 8 flights. Let $G(N, A)$ be a graph where the set of nodes N contains the set of flights X , plus 3 artificial nodes (y_a , y_b and y_c) to indicate that each of the 3 aircraft finished their scheduled flights. Each oriented arc $(i, j) \in A$ of the graph

connects two flights i and j , indicating that the ending airport of flight i coincides with the starting airport of flight j . The graph also contains oriented artificial arcs connecting all nodes i to the 3 artificial nodes. Fig. 6 illustrates this oriented graph for the illustrative example (not including the artificial nodes and arcs to avoid polluting the graph visualization) - note that this graph is not complete in the sense of having oriented arcs (i, j) and (j, i) connecting all pairs of flights i and j .

A schedule without delay ($W = 0$) for this example is illustrated in the time-space diagram of Fig. 7. The sequence of flights performed by each of the 3 aircraft in this schedule is indicated in the oriented graph in Fig. 8 with the 3 artificial nodes y_a , y_b and y_c . Aircraft a performs the sequence of flights 1, 2, 3, and 4, aircraft b performs the sequence of flights 5 and 6, and aircraft c performs the sequence of flights 7 and 8.

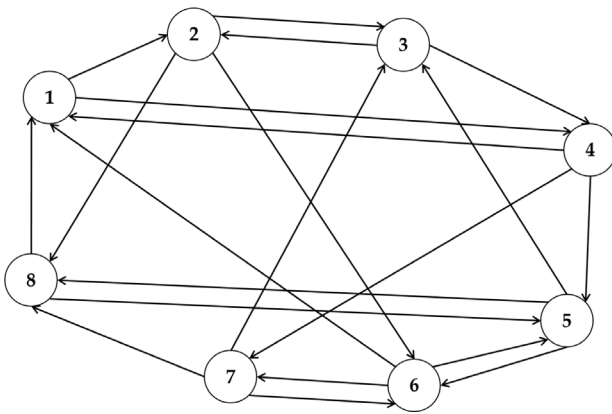


Fig. 6. Graph representing the illustrative example of Teodorović and Guberinić (1984).

Assume now that aircraft *b* presented a mechanical failure and is no longer available for its assigned flights. A minimum-weighted delay rescheduling solution for this ARP with only aircraft *a* and *c* is illustrated in the new time-space diagram of Fig. 9, with the corresponding oriented graph of Fig. 10 with only the 2 artificial nodes y_a and y_c . Aircraft *a* performs the sequence of flights 1, 2, 3, 4, 7 and 8, while aircraft *c* performs the sequence of flights 5 and 6. This solution, unlike the previous one, involves delays in the departures of flights 7 and 8 (note in Fig. 7 that the scheduled departure times of these flights are $a_7 = 12$ and $a_8 = 16$, respectively, while in Fig. 9 the rescheduled departure times become $c_7 = 17.5$ and $c_8 = 21.5$), resulting in a total weighted delay equal to $W = 302.5$. More details of this solution are described in Teodorović and Guberinić (1984).

3. Classification of problems and analysis of article content

After a quantitative survey of the SLR, the other articles published after the Teodorović and Guberinić (1984) study were reviewed and their main characteristics, approaches and contributions to the literature were highlighted. Then, in addition to the characteristics of the ARPs mentioned above and listed in Fig. 2, such as aspects about the aircraft fleet, the airport flow control and capacity limitation, the airport and aircraft operating time windows, and the optimization objectives, among others, the articles were also analyzed considering additional characteristics and practical constraints. For example, if the study considers other practical aspects motivated by real applications, such as aircraft cruising speed decisions, airport capacity constraints and flexible aircraft maintenance schedules, among others; whether the study assumes that the parameters of the problem are deterministic or uncertain (probabilistic, stochastic); if the problem decisions should consider multiple objectives and be under collaborative decision-making environments; if the ARPs are static for the purposes of day-to-day rescheduling, or if they should be considered dynamic in order to allow air operators to resolve events and disturbances as they appear throughout the day; if the ARPs appear in the context of people transportation which is observed in specific business niches such as the oil and gas industry, where a heterogeneous fleet of helicopters (i.e., a multi-fleet of different helicopter types) must meet the daily needs of offshore platforms and other maritime units, among others.

Moreover, as also considered in previous ARP surveys, the type of network representation of the problem is also identified in the present study for each reviewed article. The three most commonly used network representations are: connection networks, time-line networks, and time-band networks. A connection network is an activity-on-node network, where nodes correspond to flights and arcs correspond to

connections between flights. The oriented graph depicted in Fig. 6 is a simple example of such networks, in which nodes 2 and 6 correspond to flights 2 and 6, and the oriented arc (2, 6) indicates that flight 2 can be immediately followed by flight 6 in the route of an aircraft. A time-line network is an activity-on-arc network, where arcs correspond to aircraft activities, and each node corresponds to an event, e.g., an arrival or a departure of an aircraft from an airport. All events in an airport are located on a time line and schedule information is represented explicitly by the event nodes in these time-space networks. The time-space diagrams of Figs. 7 and 9 can be seen as parts of a time-line network by considering that the arrows indicating flights between airports in these figures are network arcs, whereas the starting and ending points of these arrows are the network event nodes along the time line. Additional oriented arcs connecting event nodes at the same airport are also part of the time-line network, for example, connecting arrival and departure times of the same aircraft at this airport. A time-band network is a network constructed dynamically as disruptions occur, composed of airport-time nodes (station-time nodes) representing activities at an airport aggregated within a certain discrete time interval (i.e., the time band), and airport-sink nodes (station-sink nodes) representing the end of the recovery period at each airport. The network arcs correspond to the flights. For more details of connecting networks, time-line networks and time-band networks, see, e.g., Clausen et al. (2010).

3.1. Single fleet

As aforementioned, Teodorović and Guberinić (1984) introduced the single aircraft type/single fleet ARP and represent it as a connected network, aiming to minimize flight delays in rescheduling. Teodorović and Stojković (1990) extended the previous model by considering airport curfews, i.e., introducing maximum time instants for flight departure and arrival at airports. Under these constraints, disturbances in the planned schedule may inexorably lead to one or more flights to be canceled. The authors model the problem as a series of shortest path problems, one for each available aircraft, which are successively solved by a heuristic based on dynamic programming. Given the limited airport working hours and the location requirements for linking individual flights, feasible paths (chains of flights) in the network have different lengths. For each aircraft, the heuristic selects the longest chain of yet unassigned flights with minimum time losses of the passengers. The flights to be canceled are those not assigned to any aircraft upon completion of the procedure. The method was used to solve an instance with 80 flights and a single fleet of 14 aircraft. In a subsequent study, the authors included constraints related to the crew, such as working hours, maximum number of takeoffs and landings, and mandatory breaks (Teodorović and Stojković, 1995).

Mathaisel (1996) described a real decision-making support system in an air operations control sector represented in a time-line network, where the out-of-kilter algorithm is used to find the minimum-cost paths in the network flow problem. The problem has a single fleet and flight cancellations are allowed, although undesirable. Cao and Kanafani (1997a,b) proposed a quadratic MIP model to minimize flight delays and cancellations, which can be extended to handle problems that allow *ferry flights*, i.e., flights to reposition an aircraft that does not contain passengers to another location where it can be used (Belobaba et al., 2015). Instead of solving the quadratic model, the authors take advantage of some properties of the latter to propose an exact iterative algorithm where a linear optimization problem is solved in two of its steps. This algorithm was applied to solve randomly generated instances with 20 to 50 airports, 30 to 150 aircraft, 5 to 12 reserve aircraft and 65 to 504 flights. Yan and Yang (1996) also represented the ARP as a time-line network and presented a mathematical model that can be extended according to the strategy used by the decision makers to recover flights: flight cancellation, flight cancellation together with

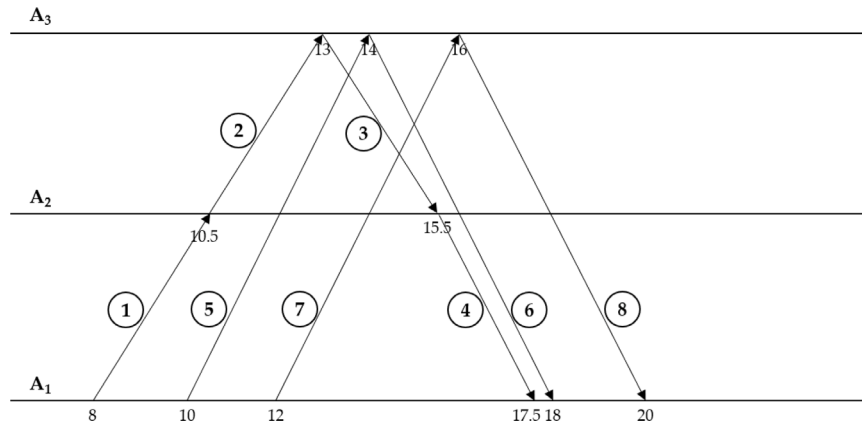


Fig. 7. Time-space diagram of the original flight schedule of the example in Teodorović and Guberinić (1984).

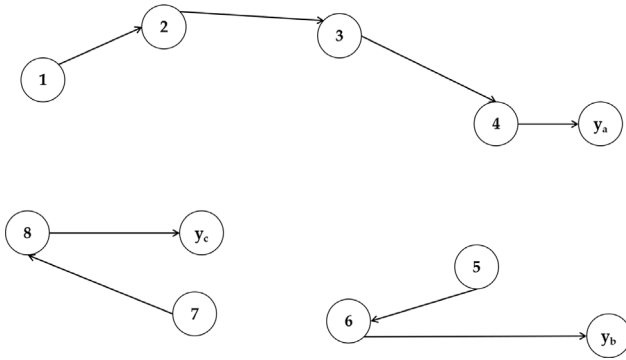


Fig. 8. Graph representing the original flight schedule of the example in Teodorović and Guberinić (1984).

ferry flights, cancellation and flight delays, or all three. The model was reformulated using Lagrangian relaxation and then solved using the subgradient algorithm. This approach has been tested with real life instances, providing solutions within 1% of difference.

Bard et al. (2001) represented the ARP as a time-band network also used in Argüello et al. (1997)'s study, partitioning continuous time into discrete intervals. Station-time nodes in this network represent activities that can be performed at each station (such as aircraft departures) during specific time intervals, and station-sink nodes depict the end of the recovery period at the stations. In each station of the space-axis, nodes are placed chronologically along the time-axis, and the inclusion of flight arcs (whose length comprise flight duration and turnaround time as well) between pairs of nodes allow the representation of all feasible flight connections in the route of each aircraft. Given the time-space network, a minimum cost flow problem formulated as a MIP model with additional constraints is then used to obtain the aircraft scheduling aiming at minimizing the sum of delay and cancellation costs. The additional constraints impose that each flight is either in a route or canceled and enforce aircraft balance at station-time and station-sink nodes (i.e., the number of aircraft at the end of a given time interval must be equal to the number of aircraft in the beginning of the interval). Flight delay costs used in the model are estimated and computed according to the delay of the arc connecting the origin and destination nodes in the time-space network. The model was fully tested with a flight schedule provided by Continental Airline comprising 30 stations, 162 flights and 27 aircraft. This strategy was further extended by Eggenberg et al. (2010) in a model called constraint-specific recovery network considering multi-fleet and maintenance operations.

In Rosenberger et al. (2003), an ARP was represented as a connection network and modeled as a set packing problem considering capacity constraints. Before solving the model, a pre-processing heuristic determines which aircraft should undergo reprogramming and rerouting. Experiments include real-life instances with 32 to 96 aircraft and 139 to 407 flights and the proposed model can be extended to minimize crew and passenger disruptions. Løve et al. (2005) proposed a multi-start local search heuristic aiming at maximizing the total revenue decremented by delay and cancellation costs, and targeting realistic sized ARPs. The local search neighborhood consists of routes resulting from a single flight swap between two aircraft located in the same airport from which the flights depart. The heuristic was further tailored to incorporate some of the scheduling rules of the British Airways and it was tested in 10 instances subjected to simulated flight delays. The flight schedules were provided by the airline and comprise an average of 80 aircraft, 44 airports and 340 flights. The study also includes an analysis of the influence of swap, cancellation and delay costs on the resulting solutions. Gao et al. (2012) dealt with an ARP whose flights have features that prioritize each other relatively, such as passenger class and flight status. Depending on the priority, different costs related to delays and cancellations are applied. The authors presented an approximate polynomial-time algorithm which was tested with a problem instance with 8 flights and 2 airports.

Kammoun et al. (2014) introduced a new approach for solving the ARP, based on discrete event systems control theory, modeling an air traffic system as a time Petri network. The main advantage of this method, compared to mathematical programming, is the possibility of modeling the position of each aircraft in a supposed air traffic network at each instant. The authors expanded the approach to deal with its inherent complexity, taking advantage of binary decision diagrams to model the network. Then, the authors introduced modifications to reduce the high computational memory cost required by the method and a genetic algorithm was introduced to optimize the time Petri networks (Kammoun et al., 2016; Kammoun and Rezg, 2018, 2019).

Wu et al. (2017a) formulated the ARP as MIP models, represented them as a distributed network in order to solve the problem rapidly, and applied a fixed-point iterative method for integer programming. The idea is to divide the solution space into independent segments and solve these with distributed computation. The motivation is the airline disruption problems caused by airports closure and the authors claim that this is the first distributed integer programming method to tackle with such large airline disruption problems. The results were promising especially for large dimension instances. The distributed computation approach was also extended to deal with multiple fleet types using a modified traveling salesman problem to generate sets of the feasible flight routes for each aircraft type, which are then assigned to the available aircraft in each fleet to obtain a recovery

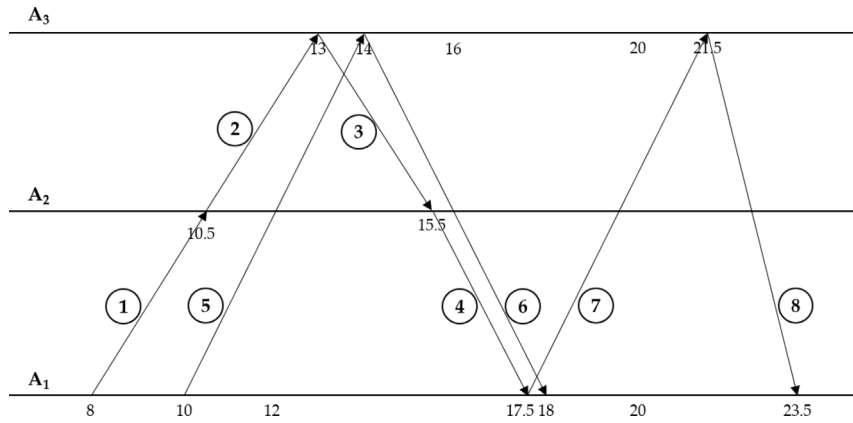


Fig. 9. Time-space diagram of the flight rescheduling of the example in Teodorović and Guberinić (1984).

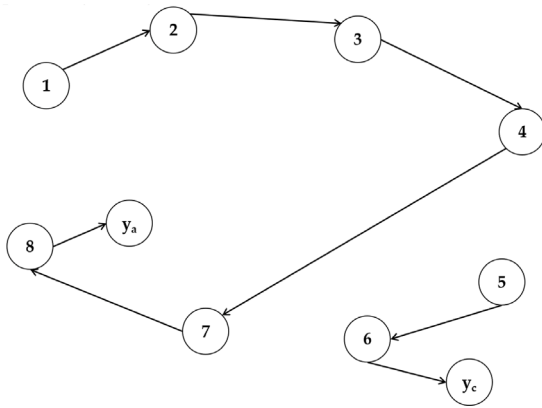


Fig. 10. Graph representing the flight rescheduling of the example in Teodorović and Guberinić (1984).

plan (Wu et al., 2017b). Wang et al. (2019) presented a simulation model based on a dynamic system framework. The model handles flight schedule recovery problems with two types of disruptions: bad weather and unexpected maintenance. Some numerical results are analyzed considering problem instances from China Eastern Airlines.

3.2. Single fleet and other practical features

Next, some studies are reviewed considering other practical characteristics of the ARP, although still considering a single fleet type. Luo and Yu (1997) sought to minimize the effects of the ground delay problem, one of the strategies used by the Federal Aviation Administration (FAA) for a more efficient use of airspace and the different capacities of airports. This strategy consists of temporarily reducing the amount of landings allowed during a certain period for certain airports, causing flight controllers to have to reschedule their previous schedules. The problem was formulated with a MIP model strengthened with valid inequalities to improve the lower bound of its linear programming relaxation, minimizing both the maximum allowable delay and the number of delayed flights. A MIP-heuristic procedure based on solving a restricted version of the model was also presented for finding good feasible solutions.

Liang et al. (2018) dealt with an ARP that simultaneously considered airport capacity constraints (i.e., maximum number of simultaneous take-offs and landings) explicitly and flexible aircraft maintenance schedules (rather than fixed, as they were generally considered). The authors developed a column-generation-based heuristic procedure that iteratively adds columns with negative reduced costs to the master

problem. The master problem with additional columns is solved and the procedure is repeated until the pricing subproblem does not result in columns with negative reduced costs. The heuristic was tested in real and simulated instances comprising 59 to 638 flights, 16 to 44 aircraft and 12 to 32 airports. Optimal solutions were found for the smallest instances and, for the largest, with a 37% gap, a satisfactory value according to the evaluated company, with computational times less than 6 min.

Erkan et al. (2019) proposed a generic mathematical model to solve an ARP under a collaborative decision environment. The ARP studied considers a single fleet and time slots at the airports. The proposed model has the least amount of essential constraints and allows stakeholders to expand it and use it for different purposes, adding new constraints, or using different objective functions according to their purposes. For validation, the authors defined a base case, with optimization objectives considered common to any stakeholder, using real data from 6 h of operation at the Minneapolis airport, USA. Applying a general purpose optimization solver with time limit of 20 min, improved solutions were obtained compared to the rescheduling performed by the airport over the same period.

In Kammoun et al. (2020), the authors proposed a mathematical model to better describe the objective and practical constraints of the ARP. An algorithm based on genetic search, similar to the previous studies in Kammoun et al. (2016) and Kammoun and Rezg (2018, 2019), is proposed as a solution method. Two of the optimization objectives are worth mentioning. Following Akturk et al. (2014) and Arkan et al. (2016), aircraft cruise speeds are decision variables that are weighed in the mathematical model in order to minimize the CO₂ emissions resulting from flights. Moreover, regarding the minimization of flights delays, two types of delays were considered: departure delay and arrival delay, which could have different values for the same flight as the speed of the aircraft is not constant.

Recently, Liu et al. (2022) presented a MIP model based on an air-rail inter-modal strategy to incorporate a high-speed rail transport mode into an aviation network for aircraft recovery. This strategy focuses on the occasional operational integration of existing airside infrastructure capacities. An inter-modal time-band network was also presented. Some numerical results were analyzed with flight schedule operation data from a Chinese airline. The approach was able to reduce the number of canceled flights and the total disruption cost. Yan and Chen (2022) presented a multiple network flow model formulated as a MIP model with certain side constraints, which aims to minimize total operating costs and deal with flight recovery and gate reassignments after disruptions. A heuristic algorithm is also proposed and some numerical results are analyzed based on problem instances of a Taiwanese airline.

3.3. ARP with uncertain parameters

The studies of the previous sections adopt the premise that all ARP parameters are deterministic. [Zhu et al. \(2015\)](#) were apparently the first to consider stochastic variables in their model for a single fleet ARP. The problem stems from the idea that, when rescheduling occurs due to the temporary infeasibility of an aircraft, such as a mechanical failure, companies generally generate their new schedules with an estimate on when the aircraft will become available again. However, the aircraft repair often takes longer than expected, so that more flights end up having to be rescheduled and, consequently, possibly delayed or canceled. The authors therefore treat such a duration as a random variable and propose a two-stage stochastic programming approach with recourse where an optimization model is presented at each stage. The first stage model was a resource assignment model on aircraft schedule recovery, aiming to minimize delays and cancellation costs. The second stage model used some simple retiming strategy to adjust the aircraft routings obtained in the first stage, aiming to minimize the expected cost on recourse decision. To solve the two-stage stochastic model efficiently, the authors designed a stochastic algorithm framework combining greedy simulated annealing ([Gao et al., 2010](#)) and a simple retiming strategy, which concerns the stochastic structure of the problem. Numerical experiments were performed in real instances with up to 23 flights and 6 aircraft, obtaining satisfactory solutions, according to the authors, in up to 5 min.

[Lee et al. \(2020\)](#) addressed the dynamic disruption management in airline networks considering both single and multiple fleet. The authors propose a jointly reactive and proactive approach that forecasts future disruptions partially and probabilistically by estimating systemic delays at hub airports, and optimizes recovery decisions in response to realized disruptions and in anticipation of future disruptions. The approach combines a stochastic queuing model of airport congestion, a flight planning tool from Boeing and a MIP model of airline disruption recovery. Recently, [Rhodes-Leader et al. \(2022\)](#) proposed a multi-fidelity modeling framework, enabling uncertain elements of the environment to be included within the decision making process. The approach uses a deterministic MIP model to find initial solutions to a single fleet ARP and a simulation optimization procedure to improve these solutions. The numerical experiments indicate that this combination is able to find good rescheduling options.

Although being also related to disruption management, research within airline schedule robustness and resilience is not covered in this survey study. This line of research makes a kind of disruption mitigation effort usually done at a tactical stage during scheduling and before disruptions are known. The goal in this proactive disruption management is to make flight schedules and aircraft rotations less sensitive to disruptions. For more details about this literature, see, e.g., [Clausen et al. \(2010\)](#).

A summary of the results observed in the review of these previous single fleet ARP studies is presented in [Table 3](#), where the articles are listed in chronological order and are compared in terms of the main characteristics of their ARPs described in [Fig. 2](#). The columns of the table summarizes these characteristics, i.e., the network representation of the problem, the main features and objectives of the problem, as well as the solution methods used. Please note in the table that some articles did not explicitly represent the problem as a network. As none of the articles reviewed in this study considered operating time windows for the aircraft, this characteristic was not included in [Tables 3](#) and [4](#).

3.4. Multiple fleet

In this section, ARP articles that consider multiple fleet of different aircraft types are reviewed. [Jarrah et al. \(1993\)](#) represented a multi-fleet ARP as a time-line network and presented two network flow

models based on the network flow model in [Busacker and Gowen \(1961\)](#). In the first model, flight delays are minimized with no cancellations allowed, whereas in the second model flight cancellations are minimized and no delays are allowed. Both models prescribe the possibility of unavailable aircraft at the beginning of the day to be put into operation later in the day, aircraft swap and the use of spare aircraft. In order to solve the models, the authors use Gowen's dual algorithm ([Busacker and Gowen, 1961](#)) for minimum-cost flow network problems which provided good solutions in short computational times for three real airport scenarios in the USA. The proposal of minimizing flight delays and cancellations independently was possibly a pragmatic choice of the authors since it was meant as a real life aiding tool, thus aligned to the flight controllers practice when dealing with aircraft shortage; the authors note that flight controllers first make use of delays, and if these are judged to be excessive, they consider canceling flights to absorb the shortage. The models were later integrated into the United Airlines' decision support system ([Rakshit et al., 1996](#)).

As aforementioned, [Argüello et al. \(1997\)](#) represented the ARP as a time-band network. The multi-fleet problem addressed encompasses several realistic characteristics simultaneously, such as: the minimum time interval between a landing and a takeoff of two different flights performed by the same aircraft (i.e., the turnaround time), airport curfews, aircraft balance at airports and mandatory maintenance schedules. The authors proposed a greedy randomized adaptive search procedure (GRASP) algorithm, tested in real instances with up to 16 aircraft and 42 flights.

[Yan and ping Tu \(1997\)](#) extended the model described earlier in [Yan and Yang \(1996\)](#) to the multi-fleet case. Later, [Thengvall et al. \(2000\)](#) presented a network flow model with side constraints based on the models of [Yan and Yang \(1996\)](#) and [Yan and ping Tu \(1997\)](#), aiming to reschedule the aircraft to minimize the deviation from the original schedule. Protection arcs and through-flights were used to encourage the routing of a flight line for each aircraft used. The solution approach was applied to instances of [Argüello et al. \(1997\)](#) and consisted of solving the linear relaxation of the model with a general-purpose optimization solver. The resulting solutions were further processed by customized heuristics until feasibility was reached. Later, the authors extended the models to consider closing hubs ([Thengvall et al., 2001, 2003](#)).

Using a connection network representation, [Andersson and Värbrand \(2004\)](#) presented a multicommodity flow model formulated as a MIP model with side constraints, which was further reformulated into a set packing model using the Dantzig–Wolfe decomposition. The recovery actions comprise flight cancellations, delays and aircraft swaps. A heuristic approach, also using Lagrangian relaxation, was applied to solve problem instances from a Swedish airline. The results revealed that it is capable to provide good quality solutions in affordable computer runtimes. In a later work, [Andersson \(2006\)](#) applied tabu search and simulated annealing to solve the problem. These heuristics use a tree-search algorithm to find new schedules for the aircraft, and a path relinking strategy to explore paths between structurally different solutions.

[Filar et al. \(2007\)](#) sought to solve the ARP from a “common good” point of view. They started from the premise that with a judicious choice of interventions (for example, propagation of delays and flight cancellations), it is possible to minimize the harmful effects of interruptions in the schedule experienced by all stakeholders: passengers, airlines, airport companies and air traffic regulatory agencies. Among its characteristics, the use of the time discretization strategy (as in [Bard et al. \(2001\)](#)) is noteworthy, and its adaptability to different situations as well. One such situation is the beginning of the day, when it is easier to modify the schedule. The other consists of emergencies, when there are even planes in the air whose landing schedules need to be modified. The model was tested in real instances at Sydney airport,

Table 3

Overview of the characteristics of the single fleet ARPs in the literature.

Article	Features				Objectives				Solution methods	
	Network ^a	ApFC ^b	ApTW ^c	RD ^d	D ^e	S ^f	C ^g	Other features and/or objectives	Heuristic	Exact ^h
Teodorović and Guberinić (1984)	Connection	–	–	–	✓	–	–	Minimum weighted delay	Heuristic with branch-and-bound	–
Teodorović and Stojković (1990)	Connection	✓	–	–	✓	–	✓	Airport curfews	Heuristic based on dynamic programming	–
Teodorović and Stojković (1995)	Connection	✓	–	–	✓	–	✓	Maximum number of takeoffs and landings	Heuristic based on dynamic programming	–
Mathaisel (1996)	Time-line	✓	–	✓	✓	–	✓	Interactive decision support system and operating cost minimization	–	Out-of-kilter network flow algorithm
Yan and Yang (1996)	Time-line	✓	–	✓	✓	✓	✓	Spare aircraft and revenue minus costs	Lagrangian relaxation algorithm	–
Cao and Kanafani (1997a,b)	Time-line	✓	–	–	✓	✓	✓	Spare aircraft and ferrying	–	Algorithm based on quadratic MIP model
Luo and Yu (1997)		✓	–	✓	–	–	–	Ground delay program, maximum allowed delay and number of delayed flights	MIP-heuristic	MIP model and GPOS
Bard et al. (2001)	Time-band	✓	–	✓	✓	–	✓	Airport curfews	Minimum cost network flow model and GPOS	–
Rosenberger et al. (2003)	Connection	✓	–	–	✓	–	✓	–	Heuristic for selecting aircraft to be rerouted	MIP model and GPOS
Løve et al. (2005)	Time-line	✓	–	✓	✓	✓	✓	Revenue minus costs	Steepest ascent local search heuristic	–
Gao et al. (2012)	–	✓	–	–	✓	–	–	Flight status and priorities	Approximate polynomial-time algorithm	–
Kammoun et al. (2014)	Time Petri	✓	–	✓	✓	–	–	Time Petri network	Genetic algorithm	Time Petri net algorithm
Zhu et al. (2015)	Time-band	✓	–	✓	✓	–	✓	Uncertain parameters and recourse costs	Stochastic greedy simulated annealing algorithm	–
Kammoun et al. (2016)	Time Petri	✓	–	✓	✓	–	–	Time Petri network	–	Time Petri net algorithm
Wu et al. (2017a)	Connection	✓	–	✓	✓	✓	✓	Airports closure	–	Fixed-point iterative method
Kammoun and Rezg (2018)	Time Petri	✓	–	–	✓	–	–	Airport curfews	Genetic algorithm	Time Petri net algorithm
Liang et al. (2018)	Connection	✓	–	✓	✓	✓	✓	Airport capacity constraints and flexible maintenance schedules	Column generation based heuristic	–
Erkan et al. (2019)	–	✓	–	✓	✓	–	–	Airport time slots and collaborative decision environment	–	MIP model and GPOS
Kammoun and Rezg (2019)	Time Petri	✓	–	–	✓	–	–	Time Petri network	Genetic algorithm	Time Petri net algorithm
Wang et al. (2019)	–	✓	–	✓	✓	–	✓	Unexpected maintenance and ferrying	Simulation model	–
Kammoun et al. (2020)	Time Petri	✓	–	–	✓	–	✓	Aircraft cruise speeds, fuel consumption and CO ₂ emissions	Genetic algorithm	–
Liu et al. (2022)	Time-band	✓	–	✓	✓	✓	✓	Air-rail inter-modal strategy	–	MIP model and GPOS
Yan and Chen (2022)	Time-line	✓	–	✓	–	–	–	Gate reassignments and operating cost minimization	Heuristic based on network flow model	–

(continued on next page)

Australia, with up to 517 flights. Eggenberg et al. (2010) represented the problem as a time-band network and presented a modeling framework that allows the consideration of operational constraints within a column generation algorithm. The problem is represented in a constraint-specific recovery network considering multi-fleet and maintenance operations.

D'Ariano et al. (2012) modeled an ARP as a job-shop scheduling problem, in which a job (in the case of the ARP, an aircraft) must perform a prescribed sequence of operations on specific machines (in the case of the ARP, airport resources), including additional real-world constraints. Two solution methods were adopted and compared. The first combines the use of a set of heuristics to determine a good

Table 3 (continued).

Article	Features				Objectives				Solution methods	
	Network ^a	ApFC ^b	ApTW ^c	RD ^d	D ^e	S ^f	C ^g	Other features and/or objectives	Heuristic	Exact ^h
Rhodes-Leader et al. (2022)	Time-line	✓	–	✓	✓	✓	✓	Uncertain parameters	MIP and simulation models	

^aNetwork representation of the problem.^bAirport flow control and/or capacity limitation.^cAirport time window.^dExperiments with real data.^eMinimizing total delay.^fMinimizing aircraft swap.^gMinimizing flight cancellation.^hGPOS = General Purpose Optimization Solver.

feasible solution, which is then used as the starting point in a routine based on a branch-and-bound algorithm. The second applies a tabu search algorithm, disturbing the pre-defined routes for each aircraft and searching for a better solution in the vicinity. Using real data from the Fiumicino airport in Rome, Italy, it was observed that the tabu search algorithm performed better. Xu and Han (2016) presented a weighted time-band approximation model formulated as a MIP model to recover flight operations and minimize delay and cancellation costs. The model was tested on problem instances based on data from China Airlines. As previously mentioned, Wu et al. (2017b) extended the distributed computation approach in Wu et al. (2017a) in order to deal with multiple fleet.

Zhang (2017) represented an ARP as a connection network and approached it by minimizing the number of aircraft routes and aiming to allocate flights to the smallest possible number of aircraft. To achieve such a goal, the author developed a method that generates possible routes from the set of flights to be retrieved, using a depth-first search algorithm and, subsequently, executing a two-stage heuristic. In the former, routes with few flights are excluded. In the latter, the remaining routes enter as a set for the solution of a MIP model that aims to allocate them to the aircraft. Two small instances (59 and 95 flights) and three large ones (417, 586 and 638 flights), both simulated, were tested, obtaining satisfactory results in up to 150 s. Lin and Wang (2018) proposed a heuristic for a multi-fleet ARP based on fast variable neighborhood search, consisting of two stages: an initialization stage and an improvement stage. In the first one, a feasible solution is sought from the initial flight plan. In the second one, aircraft swaps are carried out successively, in a greedy way, looking for better solutions up to a finite number of iterations. The method was applied to the same instances of Cao and Kanafani (1997a,b), obtaining solutions with lower total delays and consuming less computational time.

Recently, Ji et al. (2021) proposed a build-in flight feasibility verification algorithm for an ARP with flight priorities. The authors presented a model of the feasibility verification problem, which is equivalent to maximum clique formulation for networks. Numerical results were presented and analyzed based on real data gathered from a Chinese airline. Huang et al. (2022) represented an ARP as a time-line network and proposed a copy evaluation method and a solution approach for the ARP by incorporating the method within an iterative process of copy generation and filtration. The approach relies on repeatedly solving the ARP linear programming relaxation defined on time-space networks with respect to each aircraft and evaluating the copies to be added into the networks based on the solution obtained. Computational experiments using real data indicated that the approach was able to iteratively reduce the total recovery cost and lead to promising recovery schedules in affordable computer runtimes.

3.5. Multiple fleet and other practical features

Next, some studies that considered both the aircraft fleet as heterogeneous and other practical characteristics of the ARP are reviewed. Ak-turk et al. (2014) is one of the first studies to include aircraft cruise

speed decisions in an optimization model for the ARP. These decisions are not present in the studies of the previous sections, except in Kammoun et al. (2020), as cited before. While increasing cruising speed decreases travel time, therefore helping to reduce delays, this increase also raises fuel consumption in a non-linear trend. The trade-off between cruise speed and fuel consumption is significant and should be considered when aiming to minimize both the economic costs and CO₂ emissions resulting from rescheduling. The authors propose a quadratic MIP model and conclude that allowing cruising speed decisions provides better results in terms of delays and costs.

Vink et al. (2020) modeled the ARP dynamically, aiming to develop a framework that would assist air operators to resolve disruptions in the planning as these occur throughout the day. The authors developed an aircraft pre-selection heuristic that provides a set of aircraft among the total available fleet that would be more advantageous to be used in rescheduling flights. This set is used as an input for a MIP model based on parallel time-space networks, which provides a new flight schedule. The combined method was evaluated in several simulated scenarios based on real data from a US airline, comprising 600 flights, 70 airports and 100 aircraft. For these scenarios, solutions were found in less than 1 min, and the selection heuristic were able to find a globally optimal solution in 90% of the disruption instances tested. The ARP studied considers multiple fleet and the schedule recovery is limited to occur within a given time window.

Recently, Lee et al. (2022) proposed Q-learning and double Q-learning algorithms using the reinforcement learning approach for an ARP derived from a South Korean airline. They presented an artificial environment of daily flight schedules and the Markov decision process for aircraft recovery. Some numerical experiments indicated that the reinforcement learning algorithms were able to recover disrupted flight schedules, and that the approaches easily adapt to different objectives and realistic conditions. Guo et al. (2022) studied an ARP with electric propulsion vertical take-off and landing (eVTOL) aircraft in the context of airport shuttles and intercity flights in urban air mobility. A distinctive practical feature of this ARP is that the aircraft charging tasks should be optimized jointly with the aircraft routing. The authors proposed an extended MIP model based on the models in Rosenberger et al. (2003) and Liang et al. (2018), and a branch-and-price method to solve it exactly. Different numerical experiments were carried out on problem instances of Chinese airport clusters showing that the method can provide efficient and timely recovery solutions in real-world operations.

3.6. Multi-objective ARP

Most studies on the ARP reported in the literature involve only a single objective; for those involving more than one objective, most are treated as a single objective by means of a linear weighted sum objective function. The multi-objective ARP articles discussed in this section are also multi-fleet.

Table 4

Overview of the characteristics of the multi-fleet ARPs in the literature.

Article	Features				Objectives				Solution methods	
	Network ^a	ApFC ^b	ApTW ^c	RD ^d	D ^e	S ^f	C ^g	Other features and/or objectives	Heuristic	Exact ^h
Jarrah et al. (1993)	Time-line	✓	–	✓	✓	✓	✓	Spare aircraft and ferrying	–	Network flow algorithm
Argüello et al. (1997)	Time-band	✓	–	✓	✓	–	✓	Airport curfews, airport balance and maintenance schedules	GRASP	–
Yan and ping Tu (1997)	Time-line	✓	–	✓	✓	✓	✓	Revenue minus costs	Lagrangian relaxation algorithm	–
Thengvall et al. (2000)	Time-line	✓	–	✓	✓	✓	✓	Revenue minus costs	Specific heuristic	MIP model and GPOS
Thengvall et al. (2001)	Time-line	✓	–	✓	✓	✓	✓	Closing hubs and revenue minus costs	–	MIP model and GPOS
Thengvall et al. (2003)	Time-band	✓	–	✓	✓	✓	✓	Closing hubs and revenue minus costs	–	MIP model and GPOS
Andersson and Värbrand (2004)	Connection	–	–	✓	–	✓	✓		Lagrangian relaxation algorithm	
Andersson (2006)	Connection	✓	–	✓	–	✓	✓		Tabu search and simulated annealing	
Filar et al. (2007)	–	✓	–	✓	✓	–	✓	Time discretization strategy	–	MIP model and GPOS
Liu et al. (2008)	Connection	✓	–	✓	✓	✓	✓	Ground turn-around times and multiple objectives	Multi-objective genetic algorithm	
Eggenberg et al. (2010)	Time-band	✓	–	✓	✓	–	✓	Maintenance operations	Column generation algorithm	
Liu et al. (2010)	Connection	✓	–	✓	✓	✓	✓	Multiple objectives including delay time variance	Multi-objective genetic algorithm	
D'Ariano et al. (2012)	–	✓	–	✓	✓	–	–	Job-shop scheduling	Tabu search	Branch-and-bound
Akturk et al. (2014)	Time-line	✓	–	–	✓	✓	–	Cruise speed decisions and ferrying	–	Quadratic MIP model and GPOS
Xu and Han (2016)	Time-band	✓	–	✓	✓	–	✓	Reinforcement learning	Weighted time-band approximation approach and Q-learning algorithm	
Hu et al. (2017)	Connection	✓	–	✓	✓	✓	✓	Multiple objectives including maximum delay	Neighborhood search algorithm	MIP model and GPOS
Zhang (2017)	Connection	✓	–	✓	✓	✓	✓	Number of routes and aircraft balance	Specific heuristic	MIP model and GPOS
Khaled et al. (2018)	Time-line	✓	–	✓	–	✓	✓	Maintenance operations	Multi-objective LP model	
Wu et al. (2017b)	Connection	✓	–	✓	✓	✓	✓	Airports closure	–	Fixed-point iterative method
Lin and Wang (2018)	–	✓	–	✓	✓	✓	–	–	Fast variable neighborhood search	–
Shao et al. (2020)	–	✓	–	✓	✓	–	–	Risks probabilities and multiple objectives	Multi-start algorithm	–
Vink et al. (2020)	Time-line	✓	–	✓	✓	✓	✓	Maintenance operations, aircraft routes changes and operating cost minimization	Aircraft pre-selection heuristic	MIP model and GPOS
Lee et al. (2020)	Time-line	✓	–	✓	✓	–	✓	Uncertain parameters and dynamic management	Dynamic stochastic programming and queuing model	

(continued on next page)

Liu et al. (2008) and Liu et al. (2010) are examples of the first studies to address multi-objective ARPs. Both applied multi-objective genetic algorithms to find efficient solutions for daily short-haul aircraft schedule recovery problems in face of hard and soft constraints. The first study attempts to optimize objective functions involving ground turn-around times, flight connections, flight swaps, total flight delay time and a 30-minute maximum delay time of original schedules. The second considered soft constraints formulated into objectives that

contain the minimal total delay time, the minimal flight duty swap, the minimal delay time variance, the minimal number of delayed flights, and the minimal long-delayed flights.

Hu et al. (2017) is another study to address a multi-objective ARP in which three conflicting objectives are not reduced to a single objective with a linear weighted sum method. The first objective minimizes the total deviation from original flight schedules, the second objective minimizes the maximal flight delay time, and the third objective

Table 4 (continued).

Article	Features				Objectives				Solution methods	
	Network ^a	ApFC ^b	ApTW ^c	RD ^d	D ^e	S ^f	C ^g	Other features and/or objectives	Heuristic	Exact ^h
Ji et al. (2021)	–	✓	–	✓	✓	–	–	Flight priorities	Feasibility verification algorithm	
Vieira et al. (2021)	Connection	✓	✓	✓	✓	✓	✓	Flights priorities, spare aircraft, and airport capacity limitations	Specific heuristic	MIP model and GPOS
De La Vega et al. (2021)	Connection	✓	✓	✓	✓	✓	✓	Flights priorities, spare aircraft, and airport capacity limitations	MIP-heuristics	MIP model and GPOS
Lee et al. (2022)	Time-line	✓	–	✓	✓	–	–	Reinforcement learning	Q-learning algorithms and Markov decision processes	
Huang et al. (2022)	Time-line	✓	–	✓	✓	✓	✓	Maintenance operations and operating cost minimization	Iterative process of a copy evaluation method	
De La Vega et al. (2022)	Connection	✓	✓	✓	✓	✓	✓	Flights priorities, spare aircraft, and airport capacity limitations	Specific heuristic	MIP model and GPOS
Guo et al. (2022)	Connection	✓	–	✓	✓	✓	✓	eVTOL and aircraft charging tasks		MIP model and branch-and-price

^aNetwork representation of the problem.

^bAirport flow control and/or capacity limitation.

^cAirport time window.

^dExperiments with real data.

^eMinimizing total delay.

^fMinimizing aircraft swap.

^gMinimizing flight cancellation.

^hGPOS = General Purpose Optimization Solver.

minimizes the number of aircraft actually attended in swapping. Regarding the formulation itself, the authors relied on set partitioning models. For small instances, the authors developed an exact algorithm, called binary search minimum cost flow, while for larger instances, they developed a metaheuristic combining the ϵ -constraint method with neighborhood search heuristics. Small simulated instances with 5 aircraft and 24 flights and large ones with 104 aircraft and 410 flights were used for tests, resulting in satisfactory solutions in less than 1 min according to the authors. Khaled et al. (2018) presented a multi-objective MIP for the tail assignment problem, which minimizes the operation cost and the deviation from the original solution. When considering repair/recovery solutions for the initial plan under implementation, the decision maker may want to minimize operating costs, but also limit the changes with respect to the initial plan, minimize flights cancellations, among others. The ϵ -constraint method is used to generate efficient solutions for a Pareto frontier.

Shao et al. (2020) developed a multi-objective model that provides new flight schedules aiming to minimize the total flight delay and the sum of the probabilities of six operational risks, which are related to: airspace control, flight collisions, ground service, aircraft parking, ground control and taxi conflicts. Since such risks are each the result of different operational factors, their probability values act as bounds for such factors. To solve the model, the authors used a multi-start algorithm with intelligent neighborhood selection, proposed by Molina et al. (2018). The method was tested in simulated instances based on real data with 215 flights and 2 airports, and the solutions were compared to the method currently used by the operation. The method proposed by the authors was able to reduce the total delay of flights by up to 56% and risks by up to 5% of the value proposed by the current method of operation.

3.7. Other ARPs in oil and gas companies

Other recent multi-fleet ARP studies found in the literature were those by Vieira et al. (2021) and De La Vega et al. (2021, 2022),

all in the context of transporting passengers from airports to offshore platforms and other maritime units of an oil and gas company through helicopters. They consider heterogeneous helicopters and operating time windows for the airports. In the first study, Vieira et al. (2021) proposed two MIP models for ARP: one based on a flow network and another based on events. In this ARP, several characteristics were considered: the simultaneous incorporation of flights with different rescheduling priorities together with the possibility of using spare helicopters; mandatory flight precedence; the existence of a single runway for takeoff and landing at the airport; the so-called entourage flights, in which, after landing the aircraft at the destination maritime unit, they occupy it until the end of the day; strict operational rules related to daily flight schedules; and limitations on the number and type of helicopters available for some flights. In addition, the authors also proposed construction and improvement heuristics to solve the problem. For validation, both real and simulated examples were used based on real scenarios, ranging from 9 to 45 flights and 6 to 13 helicopters. Finally, solution times ranged from 0.02 s to 2.18 min.

The studies in De La Vega et al. (2021, 2022) are closely related to Vieira et al. (2021)'s. In addition to the ARP features in described above, De La Vega et al. (2021) also considered subsets of mandatory flights and constraints concerning different lease contracts of the helicopters and mandatory stops for the crew's lunch breaks. Moreover, the study presented model-based solution approaches investigating discrete-time MIP models to cope with larger problem instances, and explored ad-hoc constructive and improvement heuristics. De La Vega et al. (2022) in turn extends the continuous-time MIP models and the heuristic approaches of Vieira et al. (2021) to deal with multiple mainland airports, instead of only one, and transfers of delayed flights among these airports to the maritime units.

Similarly to the summary of the single fleet studies presented in Table 3, a summary of the results of the multi-fleet studies is presented in Table 4, where the articles are listed in chronological and are compared in terms of the main characteristics of their ARPs.

4. Insights, discussions and perspectives

4.1. Features of the ARP studies reviewed

Tables 3 and 4 show that the number of ARP studies reviewed in the literature for single fleet is almost the same as for multiple fleets. Although the single fleet consideration facilitates modeling and development of solution methods, this is not the reality of most air operators (Belobaba et al., 2015). The tables also show that most studies represent the problem as time-line networks and connection networks, followed by time-band networks, and only a small portion uses time-Petri nets (column “Network”). As for the airport flow control and operating time windows (columns “ApFC” and “ApTW”), most of the studies consider types of flow control and/or capacity limitation, for example, related to ground delay programs, simultaneous numbers of take-offs and landings, among others, while just a few studies explicitly consider time windows for airports, for example, if there are daily time windows for airport operation that depends on the sunrise and sunset times of the day. Using real data for tests is frequent, as indicated in column “RD”, with most studies considering this aspect.

As mentioned before, none of the studies consider constraints related to operating time windows for the aircraft. However, it is worth noting that the aircraft time windows can be a common characteristic in some real problems, especially when considering a longer time horizon of rescheduling, such as a week, for example. Due to issues such as aircraft and work contracts or preventive maintenance after a certain number of hours in operation, the aircraft must interrupt their operations momentarily, becoming unavailable during that period for new flights (Belobaba et al., 2015). Thus, it is important to explore formulations and methods that incorporate these characteristics to represent the problem as close as possible to the real situations.

Regarding the optimization objectives, it was observed that almost all SLR articles minimize flight delays as indicated in column “D” of Tables 3 and 4. The study in Luo and Yu (1997), for example, is an exception as the authors sought to minimize the value of the maximum allowed delay and the number of delayed flights. Regarding flight cancellations (column “C”), most studies also aim to minimize them, that is, cancellations are allowed, although undesirable. In addition to these studies, in the SLR, ARPs whose cancellations are not allowed were identified, therefore they are modeled as hard constraints of the problem, as, for example, in Kammoun and Rezg (2018) and Erkan et al. (2019). Moreover, concerning the minimization of aircraft swap (column “S”), it was observed that several studies do not cope with this objective, although this is one of the main strategies to solve the ARP in practice according to Belobaba et al. (2015). Therefore, this is yet another relevant and realistic aspect of the problem, which, although commonly encountered in practice, is not often considered in the literature. From this perspective, only half of the studies consider the three objectives simultaneously.

As it can be seen, compared to the simplicity of the pioneering model by Teodorović and Guberinić (1984) presented in Section 2.2, in the following decades, studies centered on ARP tended to involve more complex variants of the problem, and more adherent to the current practical environments, but requiring more elaborate formulations to be properly represented. These formulations involved several objectives and several additional practical constraints to be able to more accurately consider the particularities of current real problems. Column “Other features and/or objectives” presents other features and objectives considered in the reviewed studies, such as spare aircraft, aircraft ferrying, speed control, fuel consumption and CO₂ emission, maintenance operations, aircraft charging tasks (eVOLT), airport balance and curfews, maximum number of takeoffs and landings, gate reassignments, closing hubs, flights statuses and priorities, among others.

To illustrate the evolution of studies about the ARP and its solution approaches, in Appendix, the ARP based on a real application recently reported in Vieira et al. (2021) is presented. A brief comparison of this ARP with the ARP in Teodorović and Guberinić (1984) reveals that in Teodorović and Guberinić (1984) the fleet is single, while in Vieira et al. (2021) there are multiple fleets and the airport has time window constraints. Regarding optimization objectives, the model of Teodorović and Guberinić (1984) minimizes flight delays weighted by the number of passengers, while the model of Vieira et al. (2021) minimizes flight delays, aircraft changes, flight cancellations and the use of spare aircraft, simultaneously. Furthermore, it should be noted that in the latter, flight delays, cancellations and aircraft are still subdivided according to the length of the delay, type of flight canceled and type of spare aircraft used, respectively. This ARP still involves a range of constraints specific to practical applications. Most of the other ARPs published more recently in the literature also involve a diversity of realistic elements of the problems studied, requiring more elaborate formulations and they are more difficult to be solved for real problem instances, thus showing the evolution of the literature on the subject over the years.

4.2. Solution methods identified

Tables 3 and 4 also summarize the articles reviewed in the previous section according to the solution methods used to solve the ARPs. There is a slightly greater preference for the use of heuristic methods than exact methods (columns “Heuristic” and “Exact” of the table), as one would expect since real-life problem instances are often difficult to optimally solved in acceptable computer runtimes. It is also observed that some studies used both methods, either in a combined way (Zhang, 2017), sequentially (D’Ariano et al., 2012) or alternately, depending on the size of the instance (Hu et al., 2017).

As for the heuristic methods, most of the authors aimed to develop their own customized heuristics, usually with two or more stages, the first being a constructive stage (e.g., by means of greedy algorithms, multi-start algorithms such as multi-start algorithm with intelligent neighborhood selection, and others) and the second an improving stage (e.g., by means of local search algorithms, for instance, steepest ascent local search, large neighborhood search, fast variable neighborhood search, and others). Other authors applied well-known heuristics, meta-heuristics and other heuristic methods to deal with their ARPs, such as GRASP, tabu search, simulated annealing, genetic algorithms, heuristics based on dynamic programming, heuristics based on Lagrangian relaxation and subgradient optimization, matheuristics, among others. Some authors also assumed simplifying hypotheses and treated their ARPs in an approximate way, such as considering approaches based on discrete intervals of time (i.e., time slots), instead of considering the time as continuous.

Regarding the problem formulations and exact methods, most of the ARP optimization models are similar to the models of tactical scheduling problems, such as multi-commodity network flows, set partitioning and set covering problems with side constraints. The specific exact algorithms to solve these rescheduling problems are also similar to the ones of tactical scheduling problems, while the problem space is reduced by limiting the number of aircraft during the short recovery time period. Frequent use of known exact methods can be observed to solve the MIP formulations for the problem, such as branch-and-bound and branch-and-cut methods, methods based on Benders decomposition, methods based on the simplex algorithm with column generation procedures and branch-and-price methods, methods based on depth-first search methods and fixed-point iterative methods, among others. These methods were applied either using general-purpose optimization solvers (GPOS) or not. Authors have also used other approaches, such as methods based on control theory and time Petri networks, as detailed in the review of articles in previous sections.

Only a few studies explicitly considered the ARP as a multi-objective problem and applied multi-objective methods to deal with it and generate approximated Pareto frontiers, such as the ϵ -constraint method and multi-objective genetic algorithms. Furthermore, only a few studies explicitly considered the ARP parameters uncertain and analyzed the problem by means of stochastic programming approaches, robust optimization techniques, simulation–optimization procedures, as well as approaches based on Q-learning algorithms, Markov decision processes and simulation models, for example.

4.3. ARP integration with other rescheduling problems

When a disruption occurs on the day of the operation, airlines usually react by dealing with it in a sequential procedure. First, infeasibilities in the aircraft schedule are resolved by solving an ARP; then, infeasibilities in the crew schedule are resolved by solving a CRP; then, passengers reassignments are resolved by solving a PRP; and sometimes this procedure has to be iterated until a feasible reschedule is found (Clausen et al., 2010). Even though the main focus of the present survey study is on works that have treated the ARP as an individual problem and independent of other problems, the SLR also found a number of works (not reviewed in last sections and not included in Tables 3 and 4) that integrate the decisions of the ARP with decisions of other rescheduling problems. This indicates that there has been a trend in the literature towards integrating the ARP with other problems, particularly with the CRP and/or the PRP. Therefore, it is worth briefly describing some works that integrate two and more resources (e.g., aircraft, crew and passengers) in the recovery analysis (Hassan et al., 2021).

For example, Jafari and Zegordi (2010) presented an assignment model for supporting integrated decisions of the ARP and PRP, i.e., that considers the recovery of both aircraft and disrupted passengers simultaneously. The model examines possible flight retiming, aircraft swapping, over-flying, ferrying, utilization of reserve aircraft, cancellation and passenger reassignment to generate an efficient schedule recovery plan. Then, Jafari and Zegordi (2011) extended this approach by incorporating additional operational rules. Brunner (2014) expanded the solution approach to the ground delay problem developing three MIP models, also including constraints related to the passengers' itinerary and the connections between flights of the crews. The models were solved using general-purpose optimization solvers, and the results obtained from each model were compared with the solutions provided by the company.

Zhang et al. (2015) developed a two-stage heuristic to treat an ARP integrated with the CRP considering multiple fleet, maximum flight hours allowed by crew, maximum hours that the crew can stay away from their airport of origin, maximum number of landings per crew, and minimum rest between flights. They developed an algorithm that initially aims to solve the ARP exclusively and then, in the second stage, aims to meet the previously mentioned crew constraints. Hu et al. (2015) and Hu et al. (2016) modeled an ARP integrated with the PRP considering passenger transiting and using a MIP model based on set partitioning to allocate limited resources (aircraft) to activities (flights). In the first study, the network flow model considering an approximate reduced time-band network is solved with CPLEX and, in the second, a two-phase GRASP heuristic was developed in an attempt to generate feasible plans for aircraft routings and passenger reassignment. Maher (2016) studied an ARP integrated with the CRP and proposed a general framework based on column-and-row generation. This approach considers multiple secondary variables and linking constraints and was proposed as an alternative solution approach to Benders decomposition.

Jozefowicz et al. (2013), Sinclair et al. (2014, 2016) and Zhang et al. (2016) developed different heuristic approaches to solve an integrated recovery problem in the context of the ROADEF 2009 challenge.

The ROADEF challenge was proposed by the French organization of *Recherche Opérationnelle et d'Aide à la Décision* (ROADEF). It intended to bring both, the private sector closer to recent discoveries in the field of operations research and decision analysis, and the academic researchers closer to the complex problems faced by the industry. In 2009, the proposed challenge put participants in the role of an airline that, after suffering an incident that disrupted the initial flight schedule, should develop a way to reschedule them, minimizing the consequent financial costs related to the aircraft, fuel and airport usage fees, and the costs related to the passengers, such as their dissatisfaction due to delays or cancellations. In other words, the problem integrated decisions related to the PRP in the ARP. Jozefowicz et al. (2013) developed a three-phase heuristic, Sinclair et al. (2014) improved a large neighborhood search heuristic developed by the winners of the Bisailon et al. (2011) challenge, also in three phases, and Zhang et al. (2016) developed a three-stage sequential matheuristic. Regarding the results, among the studies that deal with the instances of ROADEF 2009, Zhang et al. (2016) provided the best results.

Arkan et al. (2016) and Arkan et al. (2017) developed a flight network based representation for supporting decisions on an integrated airline recovery problem that considers ARP decisions together with passenger and crew related recovery decisions. They proposed an approach that allows recovery decisions such as departure delays, aircraft and crew rerouting, passenger reaccommodation, ticket cancellations, and flight cancellations. The approach is based on the flow of each aircraft, crew member, and passenger through the flight network of the airline, and it also deals with aircraft cruise speed decisions on the flight network. Chen et al. (2020) studied an integrated problem with single fleet and without airport time windows, and proposed two formulations, one for flight scheduling and another for flight rescheduling, both combined with CRP-related objectives and constraints. Similar to Luo and Yu (1997) and unlike most other studies, Chen et al. (2020) does not minimize the total delay of flights. The optimization of the new flight schedule, regarding the flights delays, was achieved by minimizing both the maximum delay allowed and the number of delayed flights, objectives also considered before by, respectively, Luo and Yu (1997) and Hu et al. (2017). To find feasible solutions to the problem, the authors used a variation of the genetic algorithm called non-dominated sorting genetic algorithm-II proposed by Deb et al. (2002), combined with repair strategies and custom genetic operators.

Other recent studies on integrated recovery problems are found in Sun et al. (2022), Hu et al. (2021), Naz Yetimoğlu and Selim Aktürk (2021), Woo and Moon (2021), Liu (2022) and Evler et al. (2022). Sun et al. (2022) studied an integrated ARP and PRP modeled by MIP and proposed a solution approach based on a modified time-band network. They also introduced an inter-modal concept to expand airline aviation networks by including ground transportation modes. Hu et al. (2021) presented a two-objective MIP model and a heuristic with a multidirectional and stochastic variable neighborhood search algorithm for an integrated ARP and PRP considering passengers recovery with willingness (circumstance in which a passenger obeys the airline assignment). The goals are to minimize airline recovery cost and reduce passenger recovery loss. Naz Yetimoğlu and Selim Aktürk (2021) addressed aircraft and passenger recovery decisions simultaneously and proposed a matheuristic algorithm to solve the integrated problem. The objective is to produce recovery plans that maximize the airline profit while mitigating the passenger dissatisfaction, and also considering seat capacity limitations. Woo and Moon (2021) addressed an airline-driven flight rescheduling problem within a single airport, in which crew assignment decisions and ground delay programs are considered. They introduced solution approaches under uncertain parameters applying greedy heuristics and stochastic programming based on a MIP model in Brunner (2014). Liu (2022) coped with the recovery of flights disrupted by snowstorms. The studied ARP also takes into account the

passengers to take flights and the impact of passenger delay cost on the total delay time. Evler et al. (2022) dealt with the integration of aircraft routing and resource-constrained turnaround scheduling models. They proposed a rolling horizon algorithm that analyzes recovery performance over a full day of operations. The algorithm incorporates a network delay formulation with aircraft, crew, and passenger recovery decisions. The model determines the assignment of aircraft to flight routes and the allocation of scarce resources to aircraft turnarounds at the central hub airport, allowing to estimate the delay propagation in an airline network.

4.4. Future research opportunities

The analysis of the SLR indicates potential literature gaps and future research opportunities. For example, several reviewed articles in the literature consider single fleet. However, as already mentioned, in many real situations, the aircraft fleet is usually of multiple types, and airport and aircraft operating time windows can also appear, besides other specific constraints, indicating possible gaps to be better investigated. As for the optimization objectives, in most studies in this review, it is sought to minimize flight delays in the new schedule, as well as their cancellations. However, only a small portion of articles aims to minimize the swap of aircraft previously assigned to flights, although this objective can also be important in practice and points to other research opportunities. Other opportunities also appear in multi-objective ARPs in which the consideration of their conflicting objectives as a single objective by means of linear weighted sum objective functions may not be reasonable. Moreover, as pointed out in the previous subsection, the integration of the ARP decisions with recovery decisions of other resources during the day of operation, such as the ones of CRP and/or the PRP, has been increasingly active over the last years, which also represents a promising avenue for future research. Some more specific objectives were identified in certain studies, such as the aircraft balance and decisions related to the integrations of ARP, CRP and PRP.

In this context of integration of rescheduling problems, the use of algorithms based on auctions has shown promise in solving an integrated reprogramming problem for passengers and vehicles, as can be seen in Li et al. (2007a,b). The first auction-based algorithm was proposed by Bertsekas (1992). This algorithm assigns n objects to n persons in an exclusive relationship. Let a_{ij} be the benefit (or gain) of assigning object i to person j . Thus, the objective function is to maximize the total benefit. In the auction algorithm, each object j has a price p_j , and that price becomes increasingly expensive as people “bet” for their respective best object, that is, the object for which the corresponding benefit minus price is maximum. This algorithm consists of two stages: the betting stage and the designation stage. In the betting stage, each person with no object assigned bets for her/his best object. In the assignment phase, each object is assigned to the person who placed the highest bet for it as each object can have more than one bet. The algorithm ends when all objects and persons are assigned to each other.

Furthermore, the consideration of parameter uncertainties in the modeling and solution of the ARP is a field of research that is under explored. In the case of this review, only a few studies deal with the problem stochastically, although the ARP environment is commonly covered with uncertainties (Belobaba et al., 2015). To illustrate this, as exemplified by Zhu et al. (2015), it is common for authors in the literature to consider a deterministic instant of time for a given set of aircraft to be accessible to perform flights again after the event that makes them unavailable. However, such instant is only an estimate and may lead to further delays if such aircraft are not available by the specified time. For example, if a given aircraft fails at 10 am, in most ARP studies, the authors generally assume that such aircraft will be functional again after a specified number of hours (referred to as

t for illustration purposes). However, in practice, such value of t is only an estimate, so that if the aircraft is only operational again after t , further delays in the schedule may arise. Therefore, possible formulations of the problems with uncertain parameters, for example, based on stochastic programming approaches, robust optimization techniques and fuzzy sets, among others, could be considered to better handle such parameter uncertainties and obtain more robust rescheduling solutions in practice.

Another example of an ARP parameter that is uncertain in practice, but generally treated as deterministic in the literature, is the aircraft travel time, which can generate even more delays if the aircraft flies at a lower speed than expected. Aircraft speed decisions were handled in, for example, Akturk et al. (2014), Arkan et al. (2016) and Kammoun et al. (2020), but in a deterministic manner. Additionally, while the ARP usually arises after events that cause one or more resources to become unavailable, when considering a longer rescheduling horizon, such as a full day or several days, it seems convenient to also include considerations regarding the probabilities that, during this horizon, new events cause unavailability to occur, making rescheduling with dynamic considerations even more robust. Other interesting line of research consists of developing simulation models, or even simulation-optimization, to solve the ARP, approaches that are still scarce in the literature on the subject, unlike what happens with rescheduling problems of other modes of transportation (Luethi et al., 2009; Berger et al., 2011; Almodóvar and García-Ródenas, 2013).

Other opportunities for future research would be to study the ARP in the context of air cargo transportation, instead of air passenger transportation. Little work has been done on recovery problems of the air cargo business under different disruptions, for example, under cargo demand disruption (Delgado and Mora, 2021). Other approaches that have been explored more recently in different problems and that can also be useful to analyze ARPs are data-driven models, which consist of the intelligent use of data. These approaches use machine learning, big data, data mining, and data science to extract and analyze relevant and valuable information from ARPs environments that can help in the decision-making process. Finally, despite the fact that several papers reviewed in this study report the use of real data to evaluate the proposed approaches, as shown in column RD of Tables 3 and 4, there are still very few that actually reported experiences of effectively applying these approaches in real systems. Another important topic for future research would be the validation of technical developments for ARP in practical situations.

5. Conclusions

In this review, different studies were described and assessed in the literature that address the ARP since the seminal article published in the 1980s. The growing number of publications on this problem, especially in the last decade and in the last years, indicates a growing trend in research activities in this topic. The ARP is a frequent problem in companies that carry out air transport activities, and its occurrence can generate significant operational and financial impacts. This paper identified the most frequent characteristics and features considered in the ARPs described in the literature, the main decisions, objectives and constraints involved, as well as the network representations, optimization approaches and solution methods applied to address these problems. Based on an SLR method, more than 50 journal articles published until 2022 were selected and reviewed, and these studies were evaluated in terms of different practical aspects of the problem.

Some possible research gaps in the literature based on this review were also identified and possible opportunities for future research were suggested. These research opportunities can encourage collaborations and exchanges between academia and organizations, particularly airlines. It is expected that this study will be useful to researchers and

practitioners who study this topic, not only by presenting the state-of-the-art of the ARP literature, but also by pointing out research opportunities with both scientific relevance and practical problems that may be better explored. The findings of this study have to be seen in light of some limitations. Its scope and focus were only on the air mode, only on the ARP and only on the passenger transportation. The analysis of previous and current research on the ARP could and should thus be broadened by future review studies focusing on cargo transportation, on the integration of ARP with other recovery problems such as CRP, PRP and others, and on the differences and similarities of recovery problems between the air, rail, road and maritime modes.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. The ARP in Vieira et al. (2021)

An oil and gas company must carry out daily flights transporting groups of employees, via helicopters, from its aerodrome on the sea coast (airport) to its offshore platforms and other maritime units (MUs). On the way back, the same helicopters must return bringing other groups of employees from these MUs to the airport. The company's programmers pre-book several daily round-trip flights at the beginning of the week, and in the absence of unexpected events, such as weather changes or helicopter breakdowns, personnel transport must follow these flights (called table flights) previously scheduled.

However, unexpected events are common in practice and can cause delays in flight departure times, changes in the assignment of helicopters to flights and even the rescheduling of a flight for the next day (called transferred flights), therefore requiring a schedule review of previously scheduled flights. In the case of transferred flights, the flight schedules for day $d + 1$ must be revised in order to be able to

schedule flights transferred from day d . When rescheduling a flight for the following day, the following characteristics and requirements regarding the airport, maritime units, flights and helicopters must be taken into account:

- **Airport:** the airport has a single, busy runway from which only one helicopter can take off at a time. Thus, the time between departures between two consecutive flights on this runway must not be less than a certain time interval (t^a). There is a daily time window for airport operation that depends on the sunrise (tw^a) and sunset (tw^b) times of the day, and all flights must take off and land at the airport under the sunlight (i.e., within this time window). After landing at the airport, the helicopter must not take off before undergoing an inspection procedure, which corresponds to the total time of preparation of the helicopter to perform the next flight, called “time on the ground”. The minimum time interval between landing and take-off for consecutive flights using the same helicopter must not be less than a certain time interval (called turnaround time) including the landing and boarding times of passengers in the helicopter.
- **Maritime units:** the location of a maritime unit of a flight is pre-determined and known in advance. Each marine unit has a single landing and take-off helipad and therefore only one helicopter should be on the ground of that unit at a time. Due to the boarding and disembarkation operations in the unit, the time interval between landing and taking off of a helicopter must not be less than a certain time interval (t^b), during which other helicopters must not land in the unit. This also influences airport departure times and therefore the flight rescheduling operation must ensure that the time interval between the take-offs of two consecutive flights from the airport runway to the same maritime unit must avoid more than one helicopter on the ground of this unit at the same time;
- **Flights:** Each flight is defined by the destination of the maritime unit and the time of departure from the airport on the current day. A flight is represented as a simple sequence (route): airport - maritime unit - airport. Additionally, each flight has a pre-assigned group of passengers going to the maritime unit and another pre-assigned group of passengers returning from this unit in the same helicopter, which are normally related to work shifts in the unit. There may be more than one flight for each maritime unit on any given day. The travel duration of a flight depends only on the destination (maritime unit) and the helicopter (assigned to the flight). A schedule flight i must not leave the airport before its original departure time, a_i . Thus, rescheduling a schedule flight i can only postpone its departure time, c_i , but never bring it forward, i.e., $c_i \geq a_i$. If it is necessary to delay a table flight, this delay $c_i - a_i$ cannot exceed the limit d_{max} ; otherwise, that flight must be transferred to the next day and rescheduled along with the next day's schedule flights.
- **Helicopters:** The fleet is heterogeneous in terms of cruising speed, passenger capacity, flight autonomy, among other characteristics, as the helicopters have different models. Moreover, some helicopters cannot fly to some maritime units because of their distance from the airport, or because of the size of their heliports. The same helicopter may perform multiple flights in a day, as long as those flights have a number of passengers booked that does not exceed the capacity of the helicopter. The number of flights (rounds) that a helicopter can make in a day is implicitly limited by the airport time window, the duration of flights, constraints related to time between departures and arrivals, among others. In addition to the helicopters previously assigned to the scheduled flights, there are spare helicopters readily available to be used (from the fleets denominated “pool” and “spot”), if necessary, but at much higher costs.

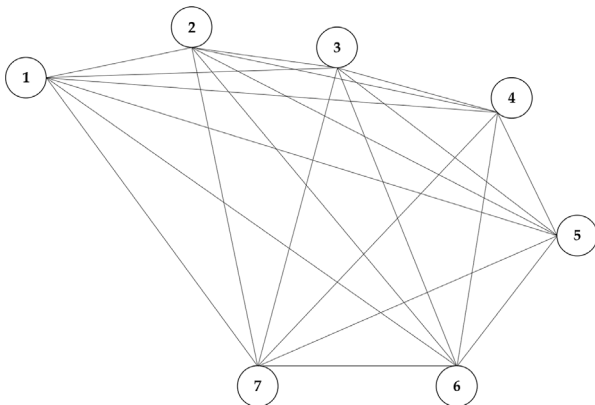


Fig. A.11. Graph with illustrative example of Vieira et al. (2021).

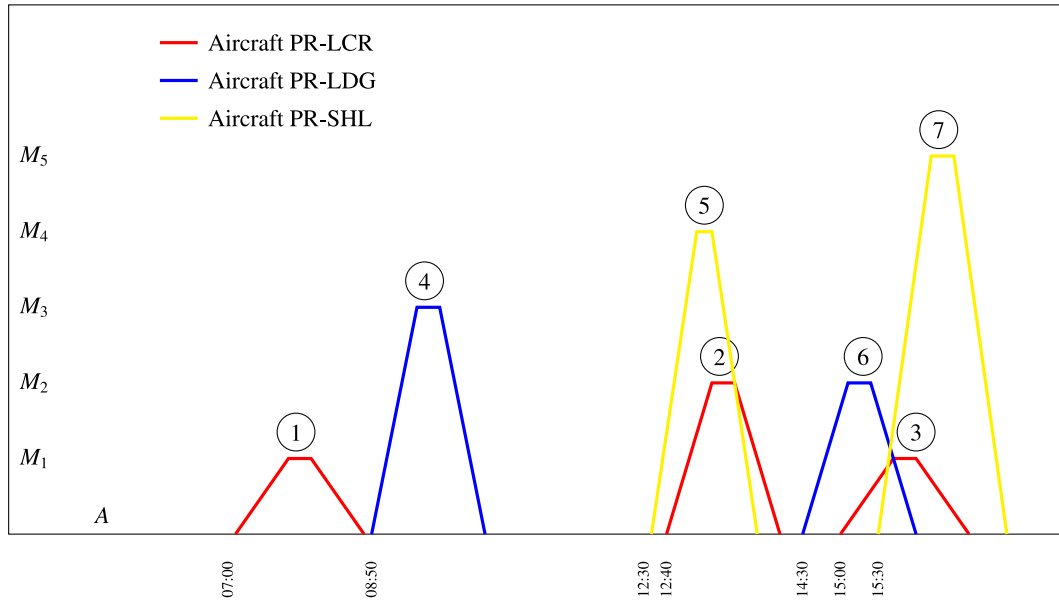


Fig. A.12. Time-space diagram illustrating the original schedule of the example of Vieira et al. (2021).

In addition to rescheduling the flights, the ARP dealt with in Vieira et al. (2021) must satisfy different practical requirements, while minimizing a weighted sum of different penalties defined by the company's operators and associated with: (i) flights transferred from the day previous and current day schedule flights that cannot be rescheduled to the current day and have to be carried over to the next day; (ii) use of more expensive helicopters than those previously assigned to flights, to cover transferred and schedule flights and possible failures of previously assigned helicopters; (iii) delays in schedule flight departure times due to rescheduling; among others. The aim is to determine a reschedule of flights that includes the maximum number of flights within the day, and if possible all flights, with the least possible penalty.

The following illustrative example, presented in Vieira et al. (2021), illustrates this ARP with an airport (A) and 5 maritime units (M_1 , M_2 , M_3 , M_4 and M_5), 7 scheduled flights between this airport and the maritime units, and 3 aircraft (a , b and c) of different types to carry out all these 7 flights. As represented in the pioneering ARP of Teodorović and Guberinić (1984), let a graph $G(N, A)$ in which the set of nodes N contains the set of flights, plus 3 artificial nodes (y_a , y_b and y_c) to indicate that each of the 3 helicopters have finished their scheduled flights. Note now that, unlike the graph in Teodorović and Guberinić (1984), this is a complete non-oriented graph (i.e., with non-oriented arcs $(i, j) \in A$ connecting all pairs of flights i and j) because two flights can be performed by the same helicopter (which is compatible for those flights) in any order from the airport. The graph also contains artificial arcs connecting all flights with the 3 artificial nodes. Strictly speaking, it would be a complete graph for each type of helicopter, containing only the nodes (flights) that this type of helicopter can perform, along with an artificial node and the corresponding artificial arcs, and the graph to represent the problem would be a composite multigraph of the union of these complete graphs for each type of aircraft. Fig. A.11 illustrates the non-oriented graph for the illustrative example (not including the artificial nodes and arcs to avoid making it difficult to visualize the graph).

A schedule without delay for this example is illustrated in the time-space diagram A.12. The sequence of flights performed by each of

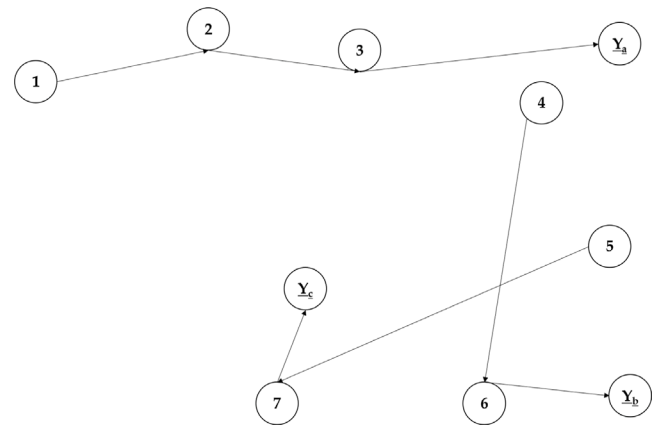


Fig. A.13. Graph with illustrative example of Vieira et al. (2021).

the 3 aircraft in this schedule is indicated in the oriented graph of Fig. A.13 with the 3 artificial nodes y_a , y_b and y_c . Aircraft a performs the sequence of flights 1, 2 and 3, aircraft b performs the sequence of flights 4 and 6, and aircraft c performs the sequence of flights 5 and 7. Let us consider now that two flights $i = 8$ and $i = 9$ were transferred from the previous day to the present day, and the other flights of the present day need to be rescheduled together. In order to solve this situation, one can resort to the use of spare helicopters, if necessary, but at a much higher cost. A rescheduling solution for this ARP now with 9 flights and the same 3 helicopters is illustrated in the new space-time diagram of Fig. A.14, with the corresponding oriented graph in Fig. A.15 with the artificial nodes y_a , y_b and y_c . Aircraft a performs the sequence of flights 1, 2 and 3, aircraft b performs the sequence of flights 9, 4 and 6, while aircraft c performs the sequence of flights 8, 5 and 7. Note that all 9 flights could be rescheduled on this day, but this solution involves penalties due to delays on flights 1, 4 and 8 (note in Fig. A.12 that the scheduled departure times of these flights are $a_1 = 7:00$, $a_4 = 8:50$ and

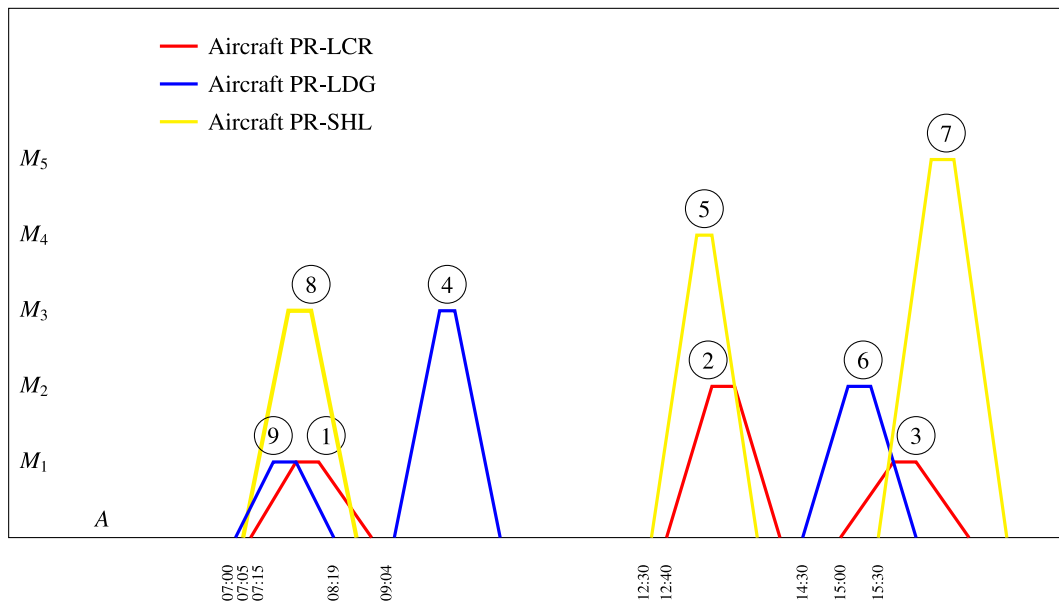


Fig. A.14. Time-space diagram illustrating the reprogramming of the example of Vieira et al. (2021).

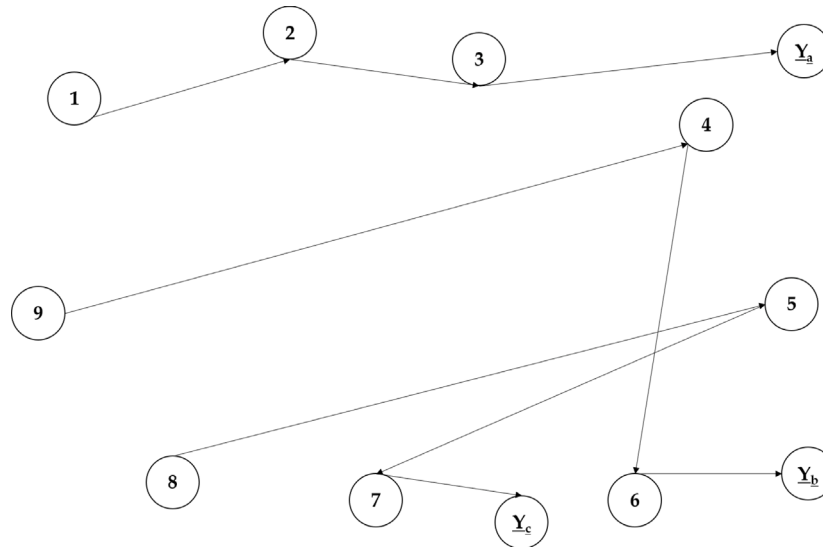


Fig. A.15. Graph illustrating the reschedule of the example of Vieira et al. (2021).

$a_8 = 7:00$, respectively, while in Fig. A.14 the rescheduled departure times become $c_1 = 7:15$, $c_4 = 9:04$ and $c_8 = 7:05$). More details of this solution are described in Vieira et al. (2021). Although not illustrated in this example, the same approach proposed in Vieira et al. (2021) may also be applied if one of the helicopters previously scheduled to perform the flights shows any problem and is no longer available for operation, as shown in the illustrative example of Teodorović and Guberinić (1984), presented earlier in the Section 2.2.

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